Mobile HCI

Lecture 1: Understanding Mobile HCI as a research field

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Human-Centered Computing (HCC), Department of Computer Science

Argentina
2008-2012
Software Engineer

France
2013-2015
Tech Entrepreneur

France
2015-2018
PhD

Japan
2019
Postdoc

Denmark
2020 - 2023
Postdoc

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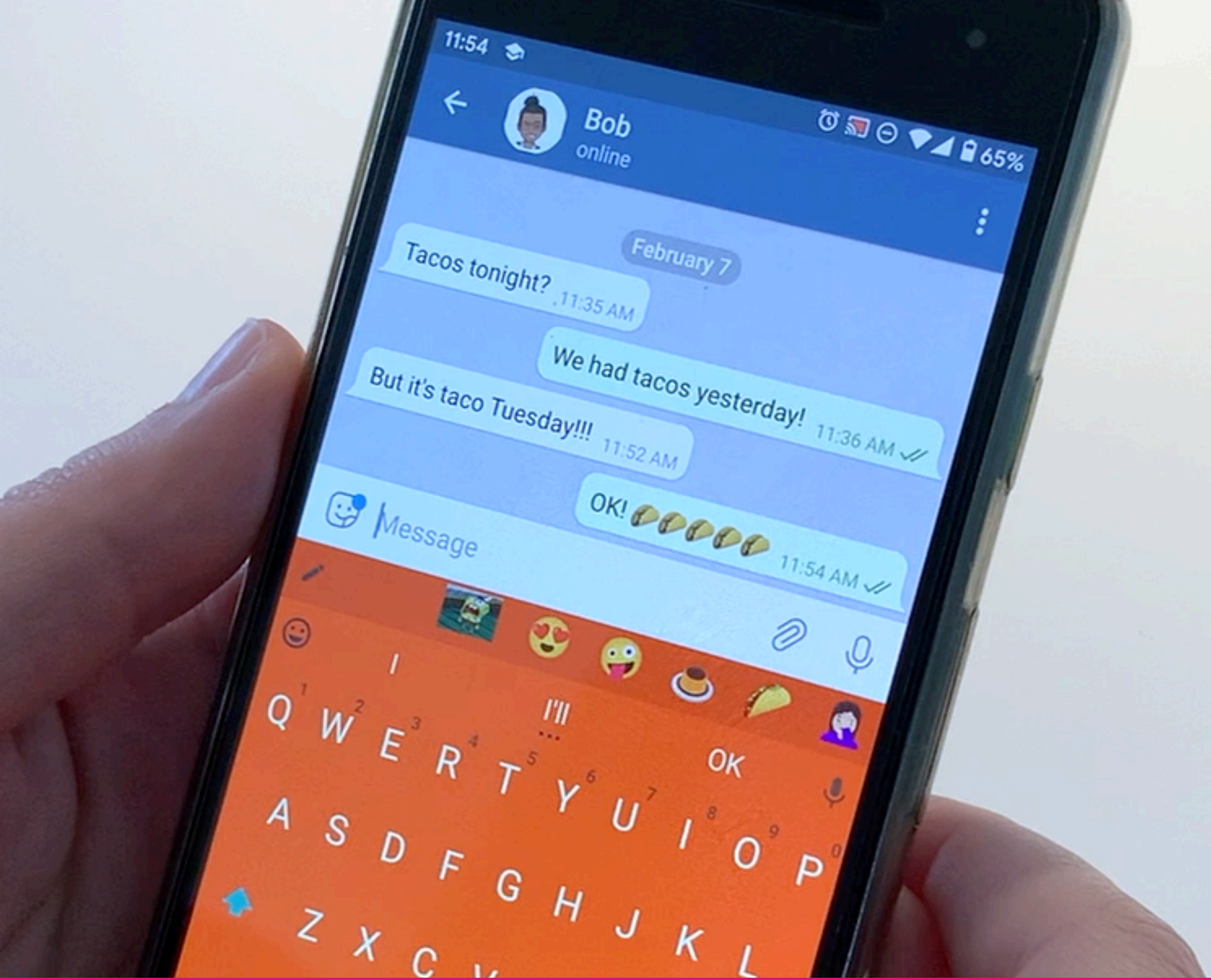
Now
Tenure-track Assistant Prof.

I study and build
communication software to grant users more control
over how they manage their relationships, expression and privacy online

Most research and innovation focus on how to design a single communication app

What are the implications of using ecosystems of communication apps and how do we design for them?



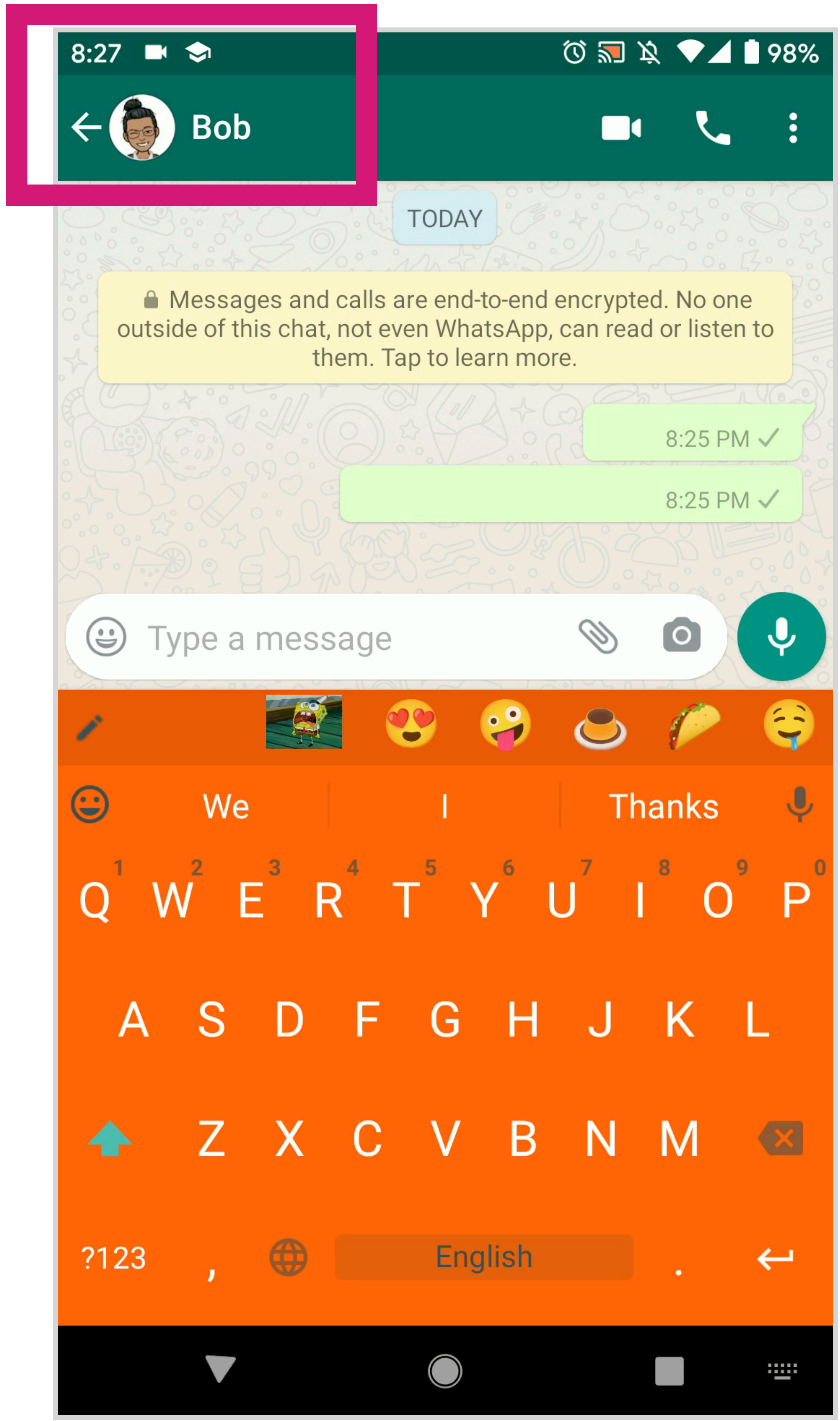
Dearboard

A co-customizable keyboard for close relationships

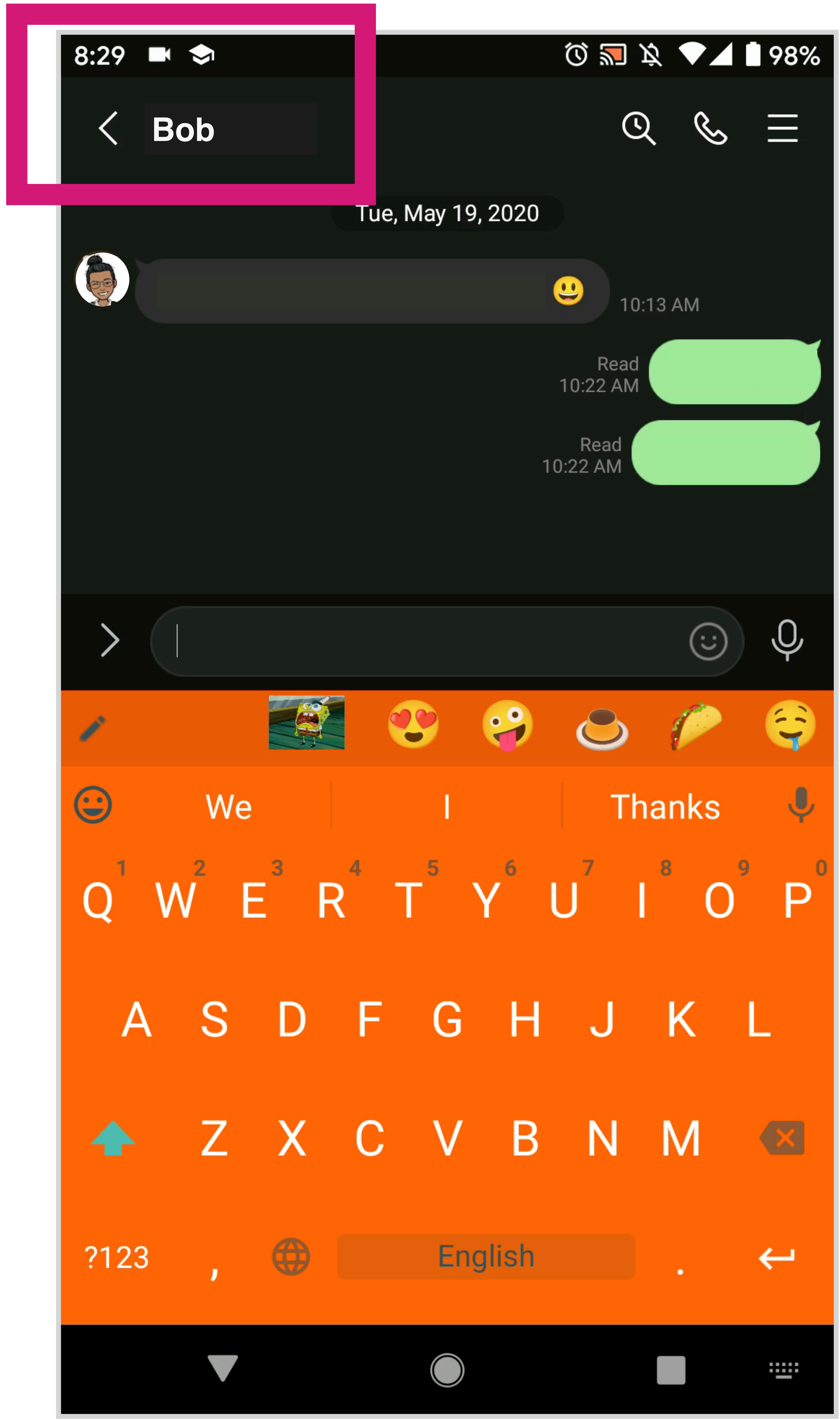
Carla F. Griggio, Arissa Sato, Wendy Mackay, and Koji Yatani

Mediating Intimacy with DearBoard: a Co-Customizable Keyboard for Everyday Messaging
CHI '21

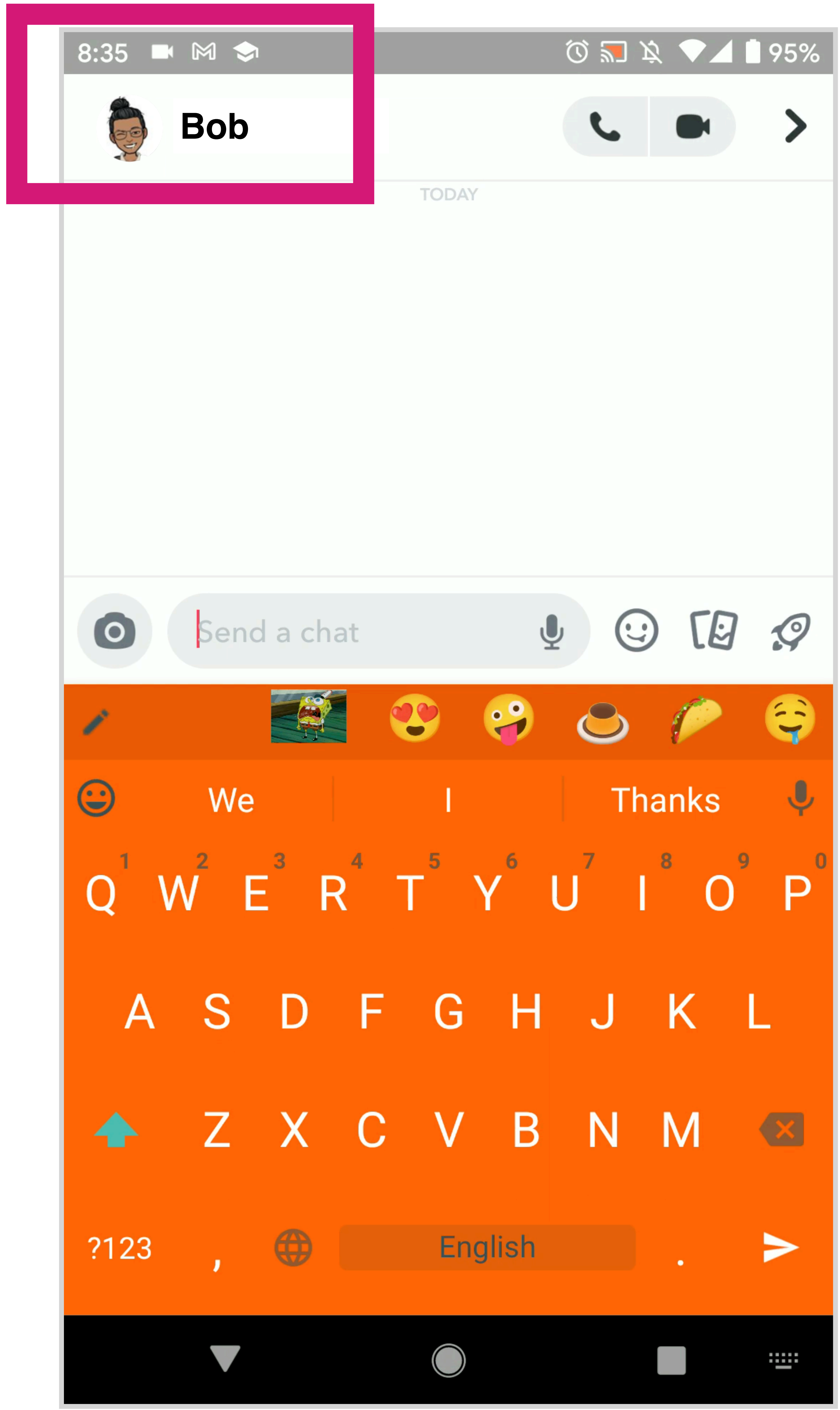

color theme
(background + keys)
shortcuts to
emojis and GIFs
Bob
online
We had tacos yesterday!
11:36 AM
But it's taco Tuesday!!!
11:52 AM
OK!
11:54 AM
We can order from La Neta this time
11:55 AM
Message
63%



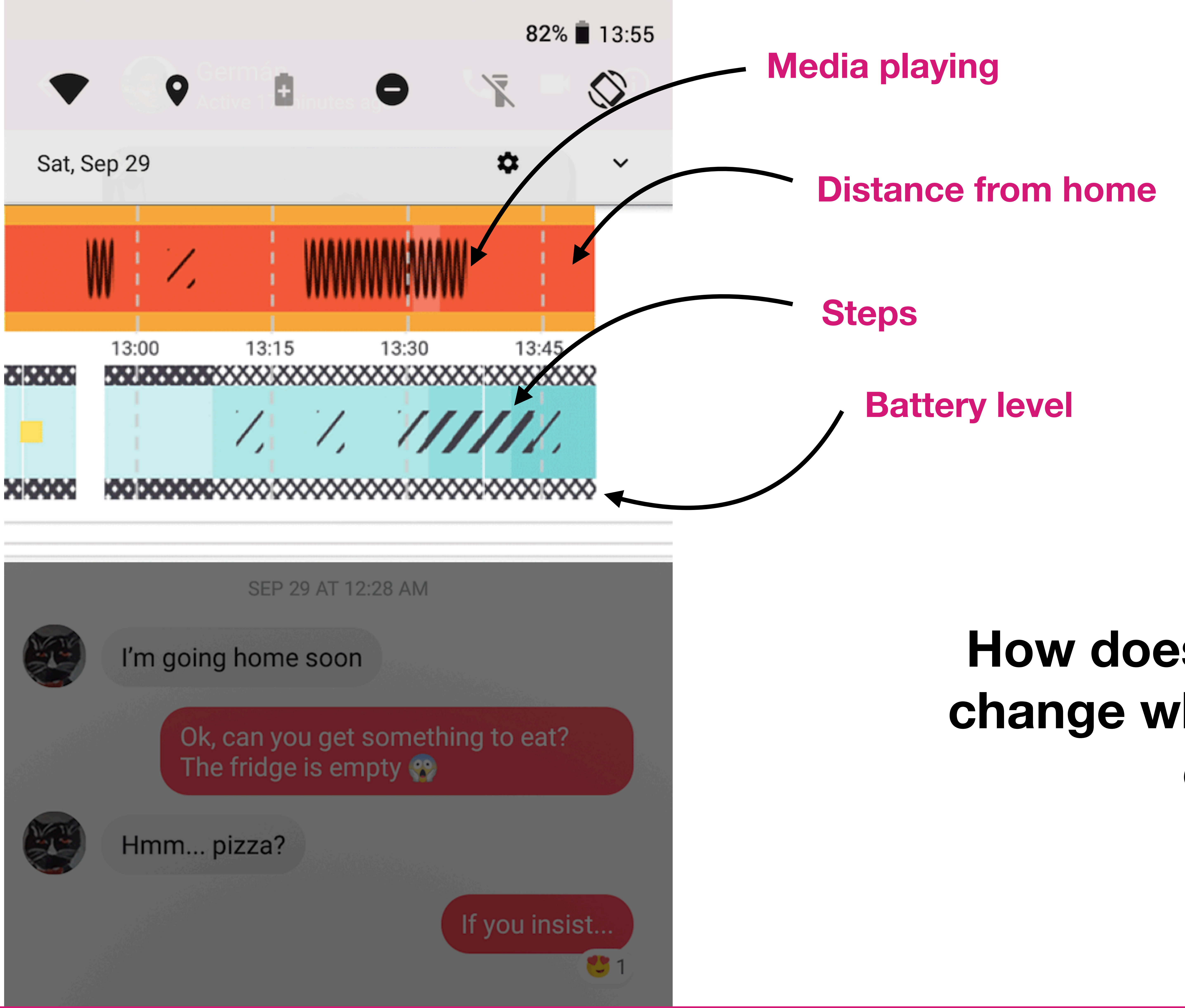
Bob on WhatsApp



Bob on LINE



Bob on Snapchat



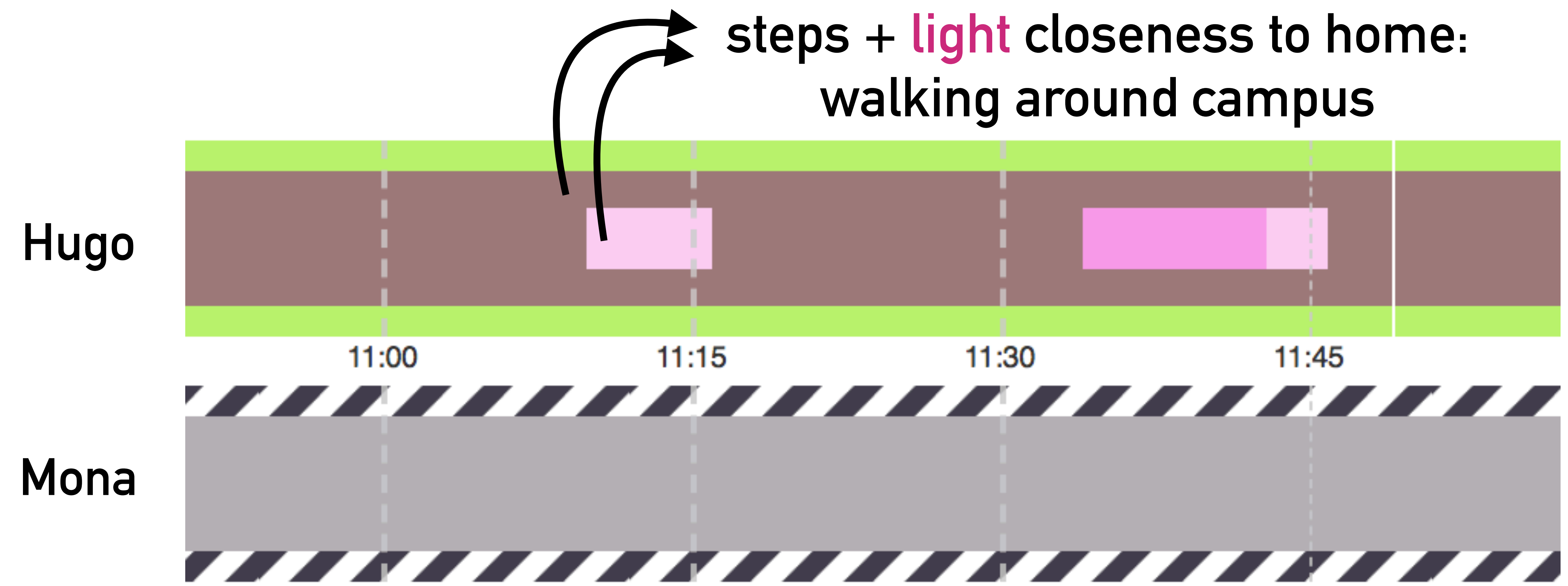
Lifelines

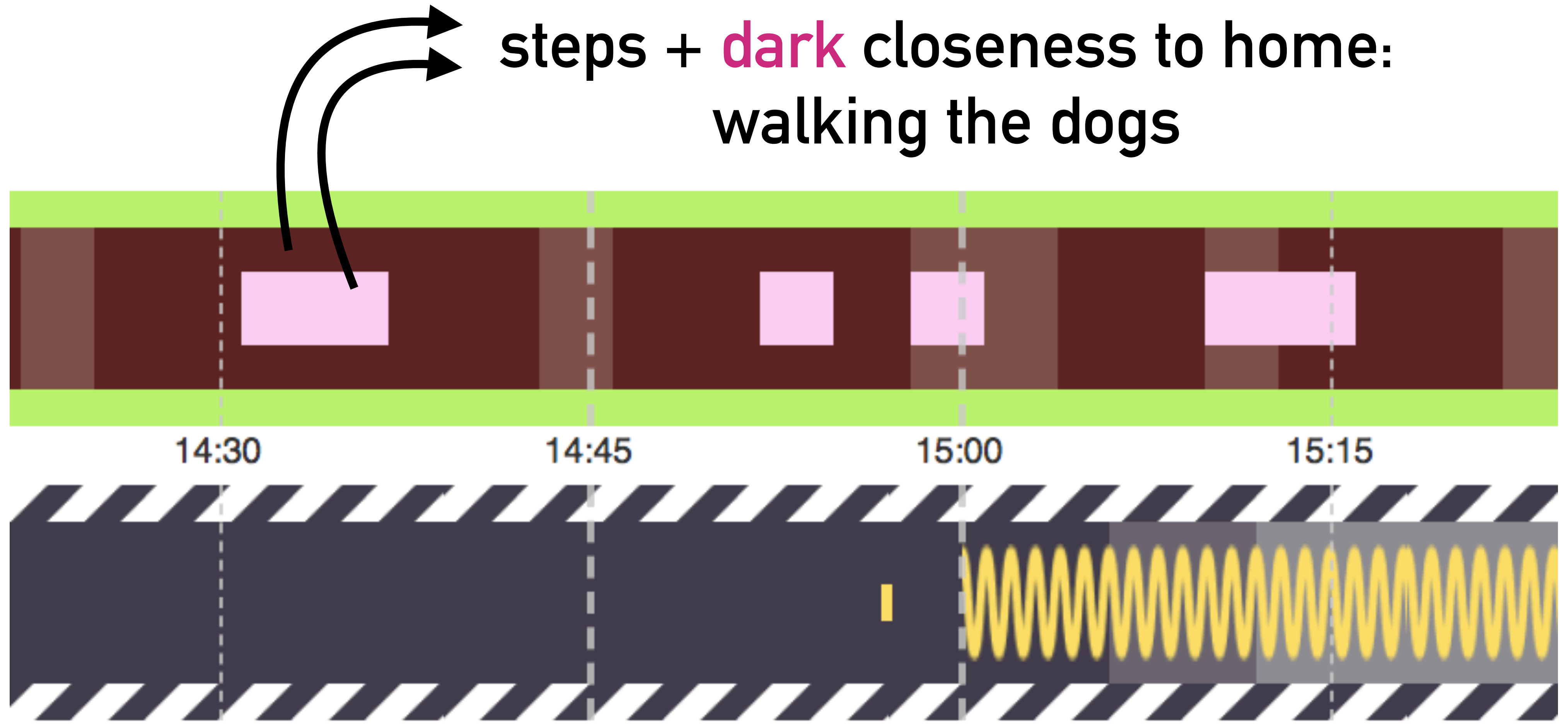
A shared visualization of mobile context for couples

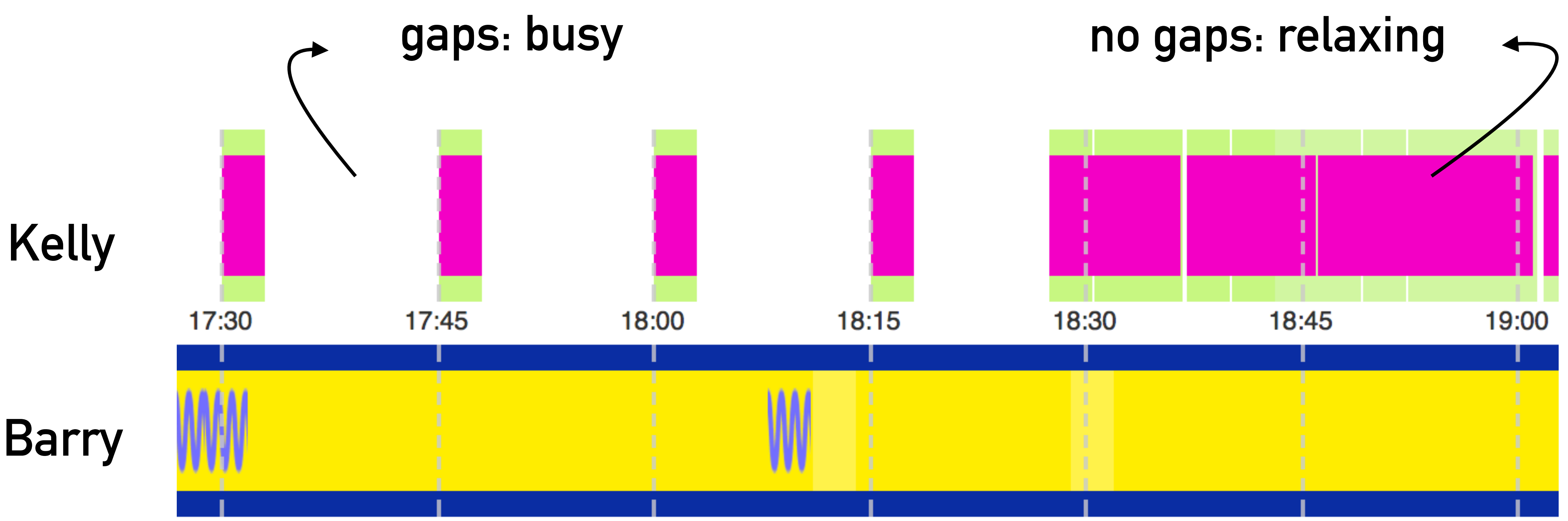
How does couple's online communication change when sharing visualizations of their contextual information?

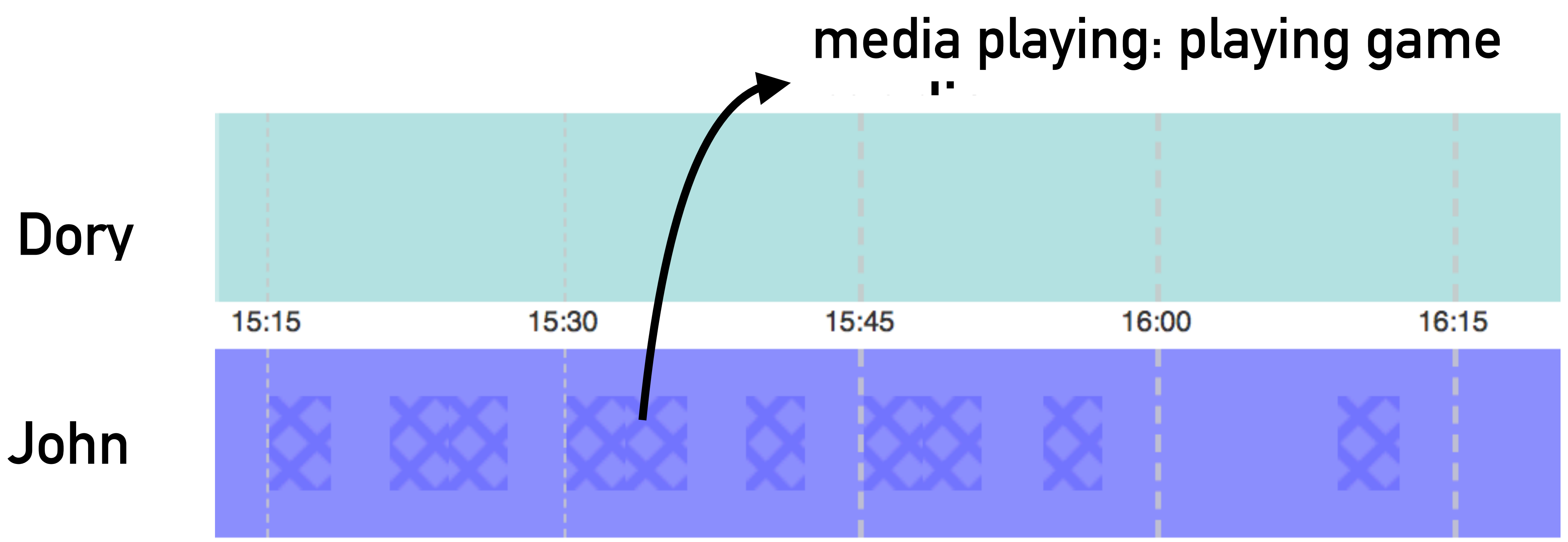
Carla Griggio, Nouwens, McGrenere and Mackay.
Augmenting Couples' Communication with Lifelines: Shared Timelines of Mixed Contextual Information
CHI '19

User study: One-month field study with 9 couples, using the system “Lifelines" on their own smartphones









Why are you interested in this course?
What do you expect to learn?

Study regulation for mHCI

KNOWLEDGE

- The student must gain knowledge of challenges and opportunities in user interaction with mobile devices, systems and services, or interaction in mobile contexts. This includes focus on technology, interaction, and usage context. In relation to this, the student must gain knowledge about the design of interaction, and the evaluation of usability and user experience.

Possible topics include:

- interaction design for small displays
- interaction design for small physical devices
- interaction design for mobile use, and dynamic usage
- contexts multi- and / or cross-device interaction
- digital ecologies
- wearables
- methods / techniques for evaluating mobile usability of user experience
- methods / techniques simulation of mobility and context
- advantages and disadvantages of resp. lab and field evaluation
- longitudinal field studies, experimental control and ecological validity

Carla's flavor of mHCI:
We'll touch on most of these these topics, but more from a **critical, academic perspective** rather than from a "practice" perspective

Additional topics:
Privacy, Communication, Accessibility, Qualitative and Quantitative Data Analysis

Study regulation for mHCI

SKILLS

- be able to design and **empirically evaluate** interaction design / user interfaces for mobile devices, systems and services.
- be able to select and apply appropriate mobile technologies **for a given context of use**, select and apply appropriate interaction techniques
- be able to select and apply **appropriate evaluation methods**.

COMPETENCES

Master principles of interaction design and **evaluation for mobile devices**, systems and services and be able to select and apply these in mobile usage scenarios

mHCI course structure

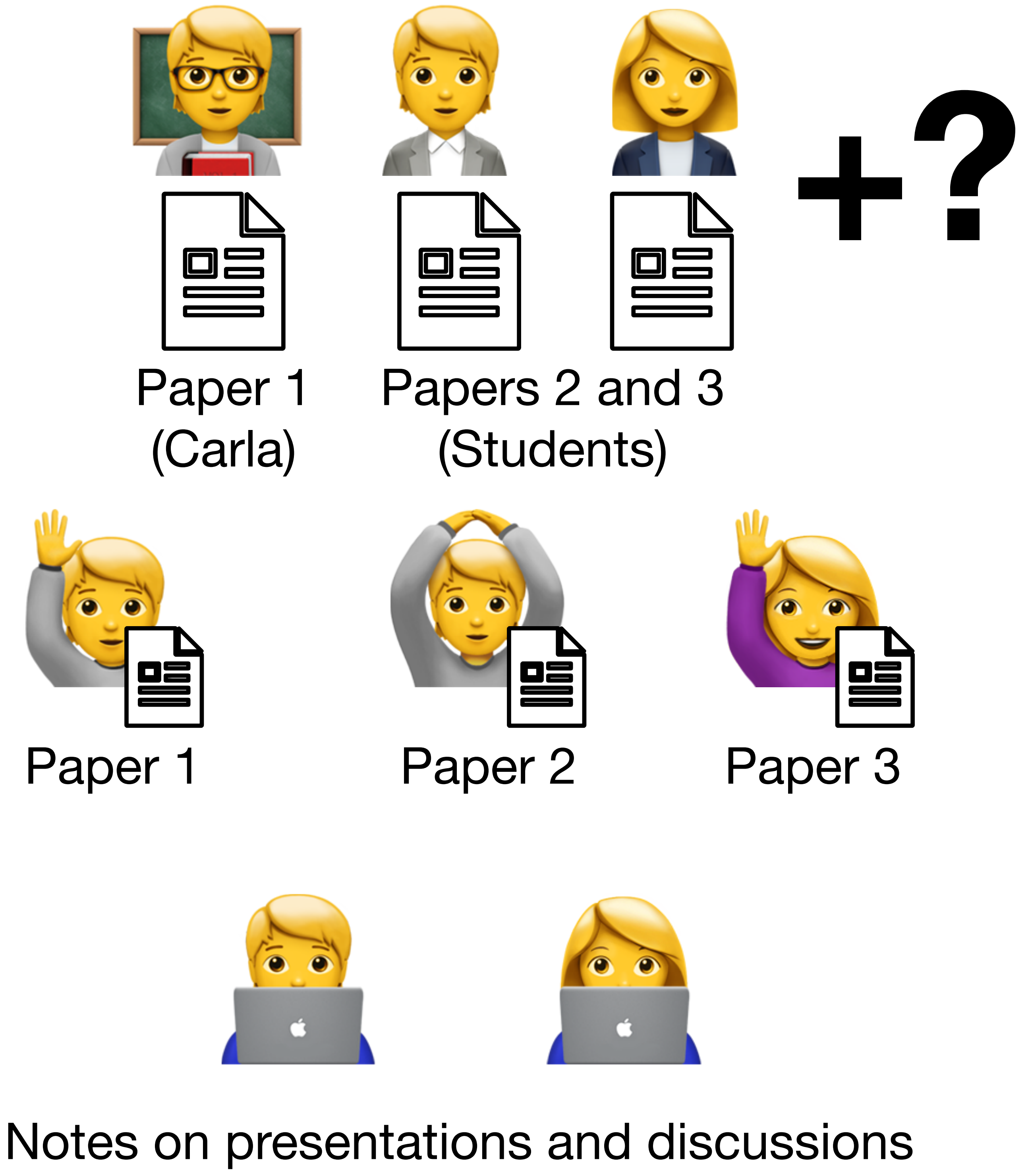
- **Part 1** (intensive, twice per week): **Mobile HCI starter kit!**
  - Intro to Mobile HCI as a research field
  - Interaction techniques and design principles
  - Designing for different contexts
  - Advanced evaluation methods for interactive systems
- **Part 2** (once per week): **Deep dive research seminars (*you* present papers! Like a mini conference)**
  - Discussions of research papers with multiple examples of the techniques, principles, contexts and methods from Part 1
  - More detailed concepts and methods
  - We take a *critical perspective* to analyze trade-offs and propose alternatives
  - **If you choose mHCI as your elective course, you must be willing to participate in these seminars as a presenter, discussant or notetaker.** You can volunteer for your preferred role and date, but all roles must be covered for all dates and I may assign students myself if needed.
- **Part 3** (last 2 lectures): **Data analysis workshops**
  - Overview of data analysis resources and methods for qualitative and quantitative data
  - Turning data into insights and opportunities for design

Mid March?
Or end of April?

Lecture structure mHCI Part 2

First half: research seminar (1:30 hours)

- **2 students + me** present an analysis of the evaluation methods from 3 research papers (1 each). These presentations will be run as a panel:
  - **3 presenters** will give a 10’ presentation of a paper, based on predefined guidelines
  - **2 discussants** will lead a Q&A session after the presentations
  - **2 notetakers** are in charge of documenting the discussion
- Seminars conclude with a short summary lecture about specific aspects of evaluation methods



Second half: exam preparation / user study workshop (2 hours)

- Guided exercise for the design of a study for your semester project, **and/or**
- Guided exercise for preparing for your mHCI exam

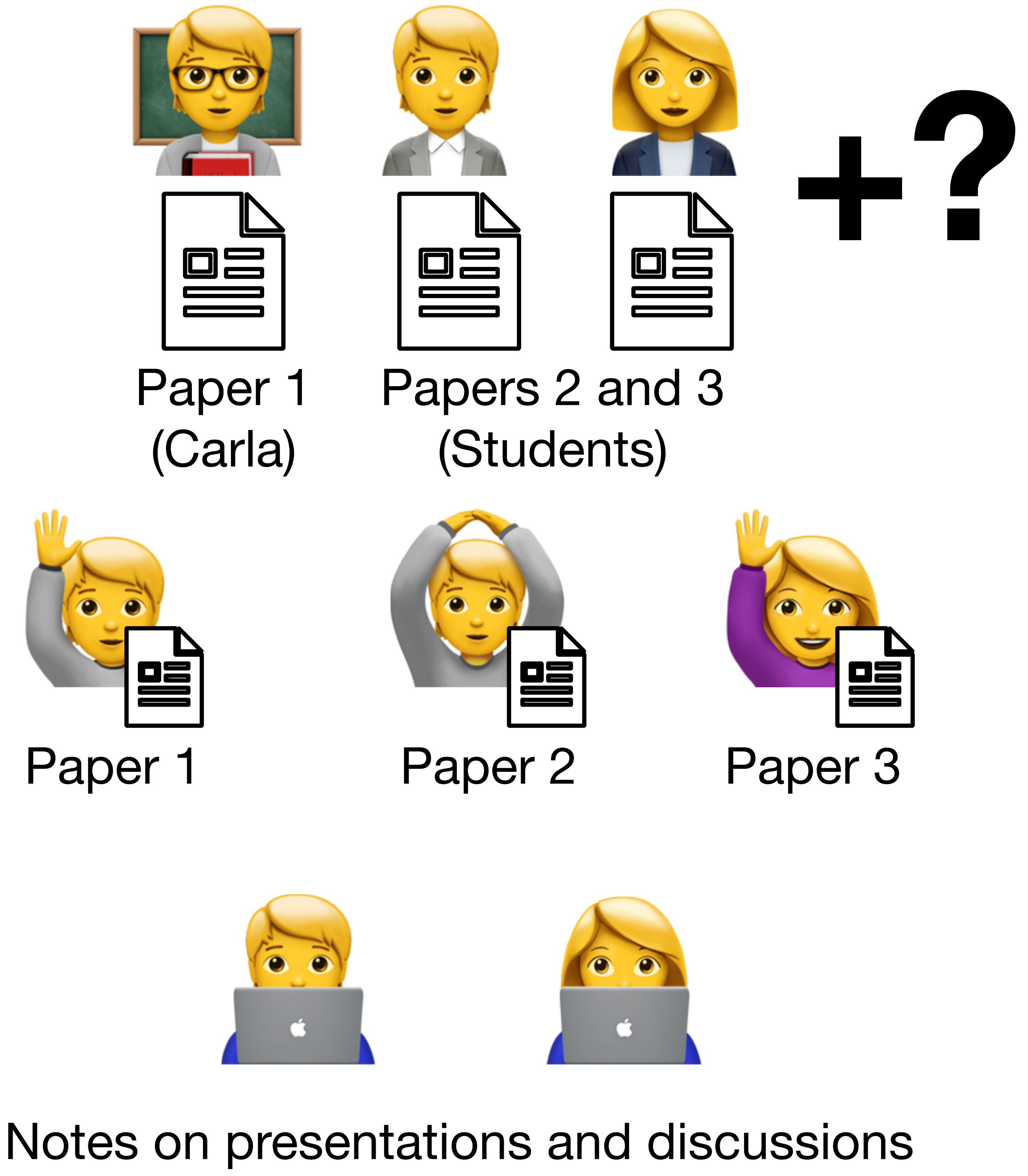
Self-study session before Part 2 starts

You will have a dedicated self-study session for reading your assigned paper and preparing for the seminar (whether it's as a presenter, discussant or note-taker).

This self-study session will be placed between Part 1 and Part 2 of the course.

The outcome of this self-study session is also your first draft for the exam presentation! :D

Every lecture after this will include a guided exercise to help you complete your exam preparation.



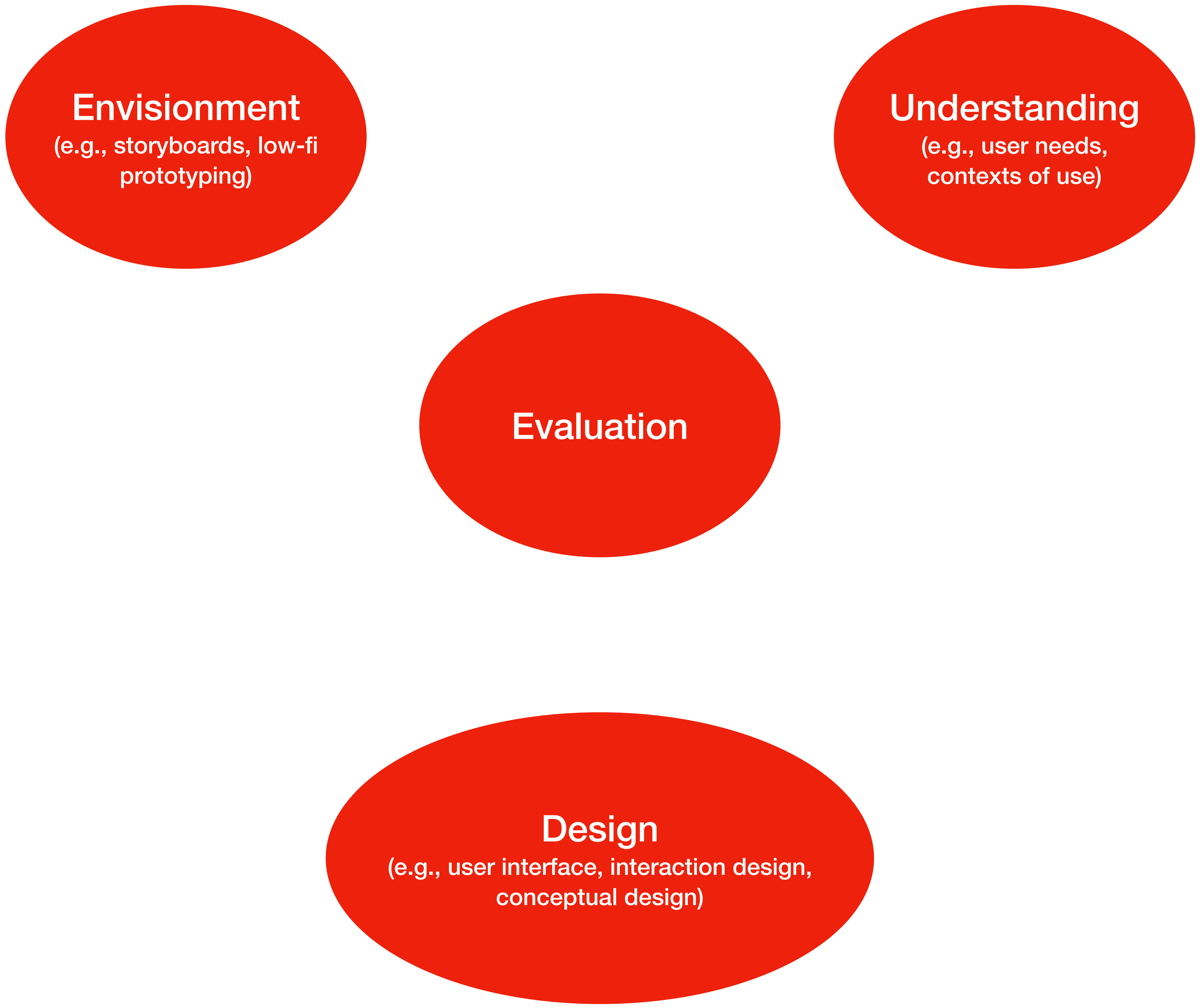
Next lectures:

- **Lecture 2: Design Principles and Mobile Interaction Techniques**
    - Bidding for preferences on seminar papers and roles start
- **Lecture 3: Context and Field Studies**
    - Bidding for preferences on seminar papers and roles ends. Paper and role assignments follow shortly.
- **Lecture 4: Overview of other research methods + Experiments**
- **Lecture 5: Self-study session**
- **Lecture 6: Digital Communication**
- **Lecture 7: Digital Ecosystems**
- **Lecture 8: Privacy**
- **Lecture 9: Accessibility and Health**
- **Lecture 10: Quantitative Data Analysis**
- **Lecture 11: Qualitative Data Analysis**
- **Lecture 12: Wrap up and Exam Rehearsal (optional)**

Goals of today's lecture

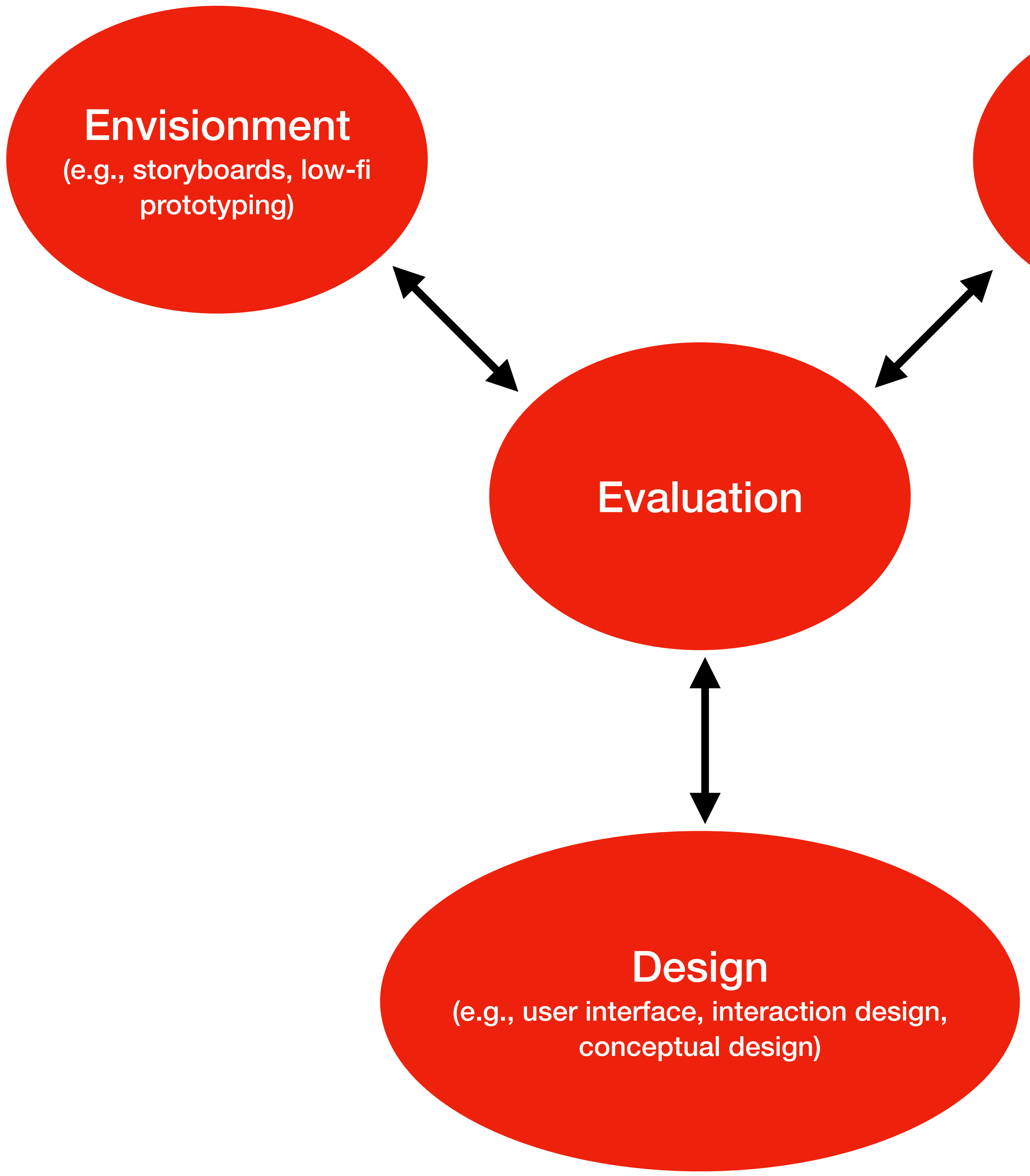
- What is Mobile HCI? Understanding the difference between *interaction design* and (Mobile) HCI as a *research field*.
- Overview of (Mobile) HCI's types of contributions with examples
- Evolving research concerns through time: Three waves of HCI research
- What are the challenges of *mobile* interaction?

Human-centred design process of interactive systems



David Benyon, 3rd edition, "Designing User Experience"

Human-centred design process of interactive systems



David Benyon, 3rd edition, "Designing User Experience"

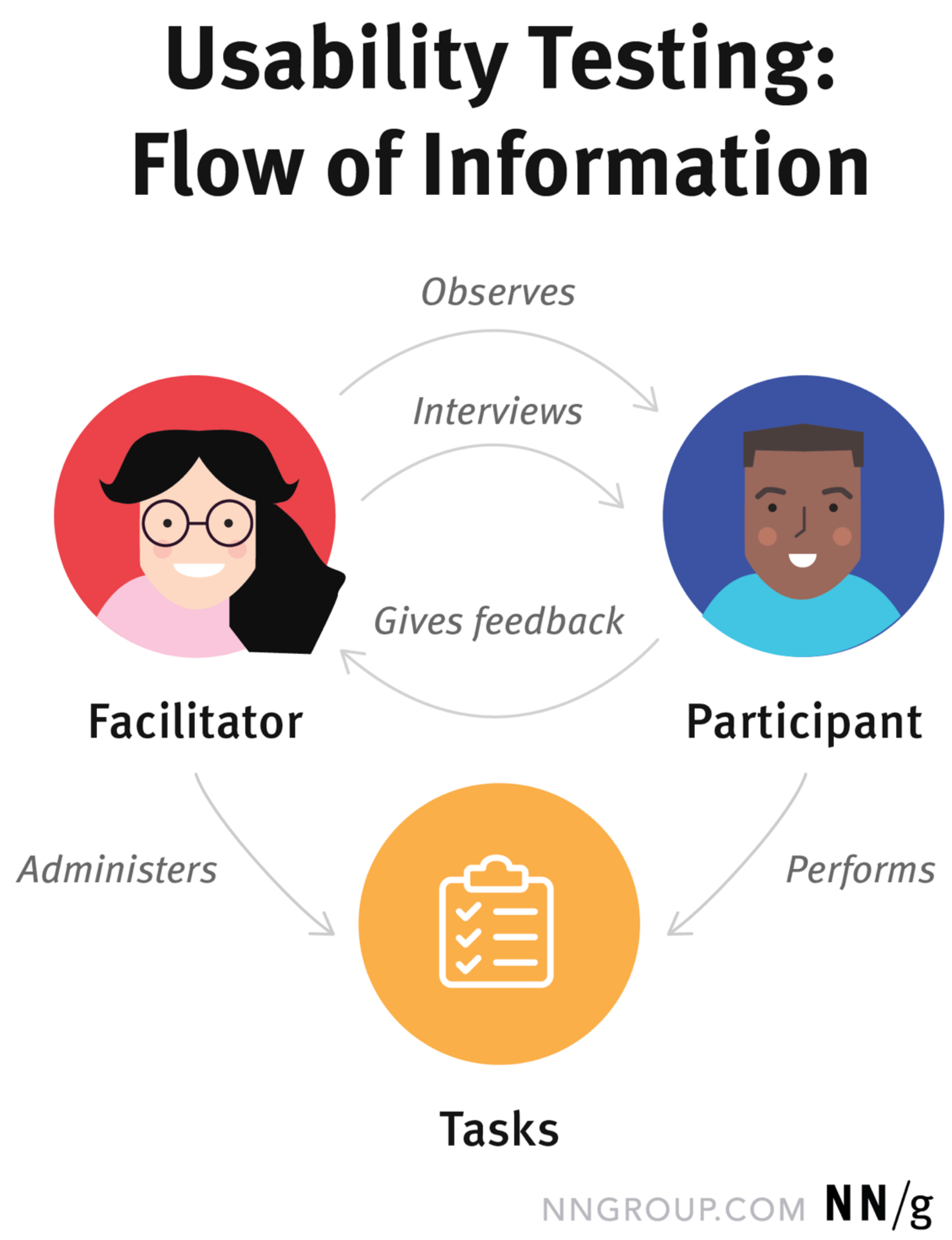
Evaluating the usability of a system

*Usability: "The **effectiveness**, **efficiency**, and **satisfaction** with which specified **users** achieve specified **goals** in **particular environments**."*

Effectiveness: the accuracy and completeness with which specified users can achieve specified goals in particular environments

Efficiency: the resources expended in relation to the accuracy and completeness of goals achieved

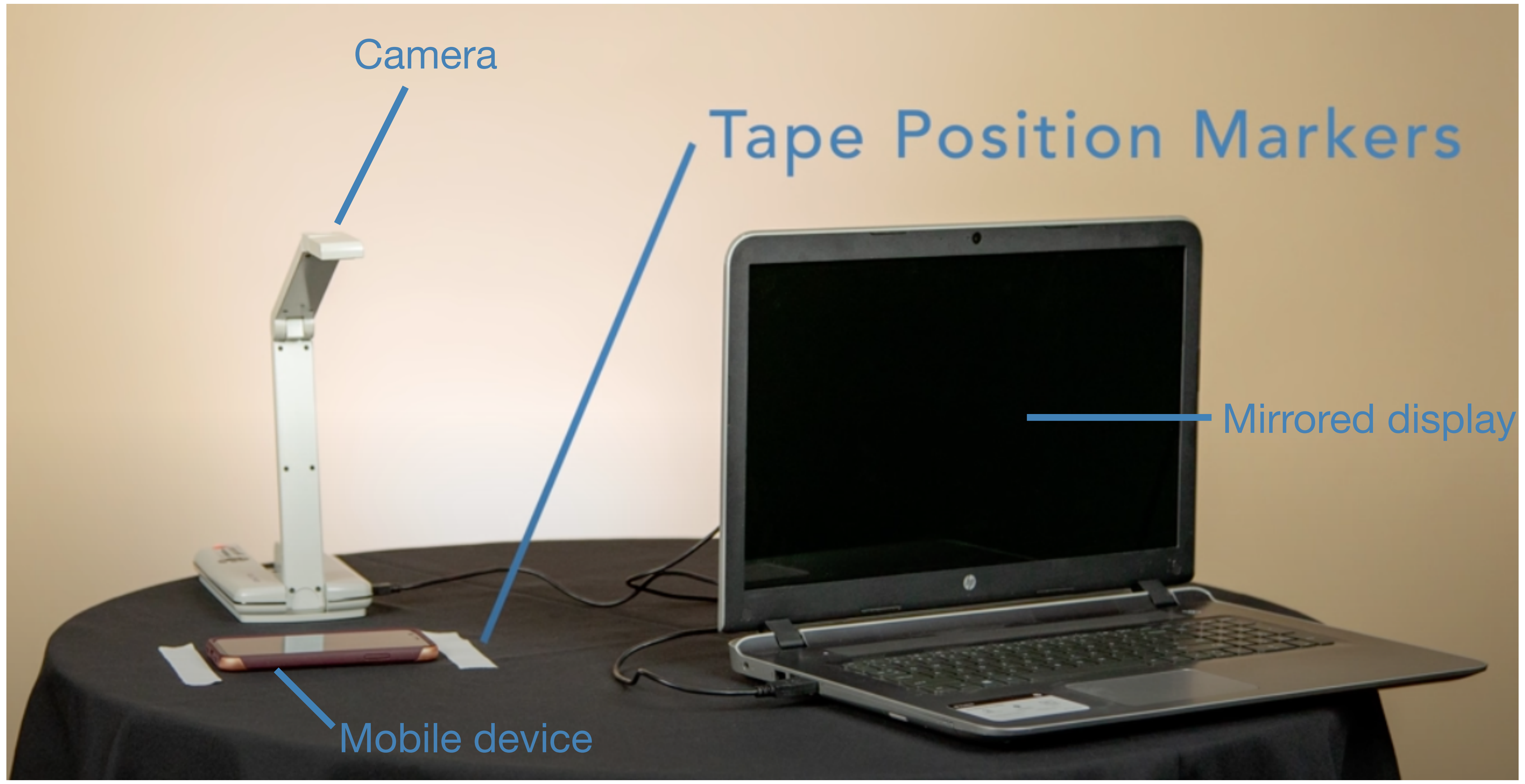
Satisfaction: the comfort and acceptability of the work system to its users and other people affected by its use



Evaluating the usability of a system

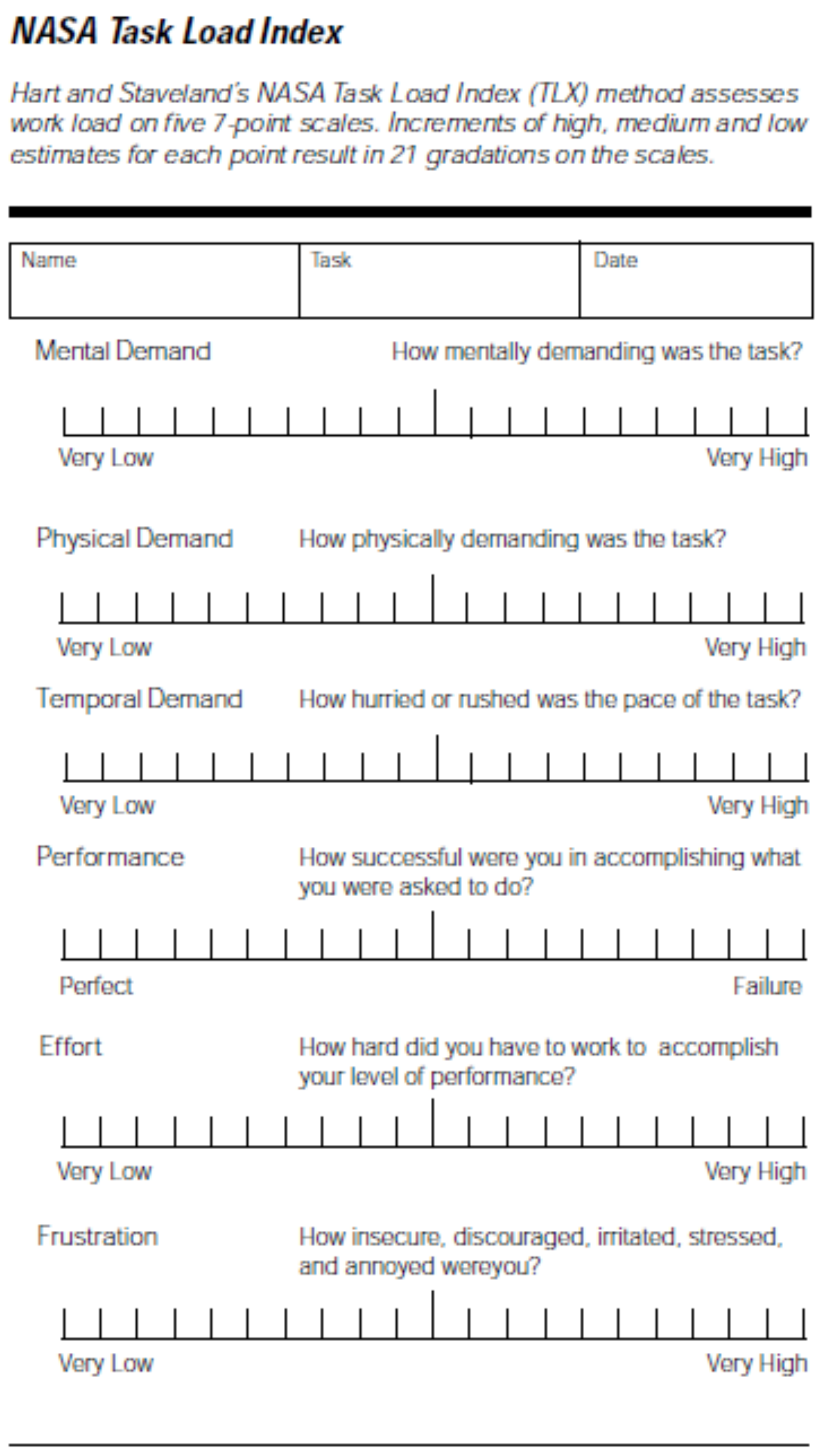
Empirical methods

(collecting and analyzing observations or measurements of real phenomena, e.g., how users behave or feel)



https://www.youtube.com/watch?v=NdqTbpyvaIg

Usability study with Think Aloud Protocol



NASA Task Load Index

Hart and Staveland's NASA Task Load Index (TLX) method assesses work load on five 7-point scales. Increments of high, medium and low estimates for each point result in 21 gradations on the scales.

Name	Task	Date

Mental Demand — How mentally demanding was the task?
Very Low — Very High

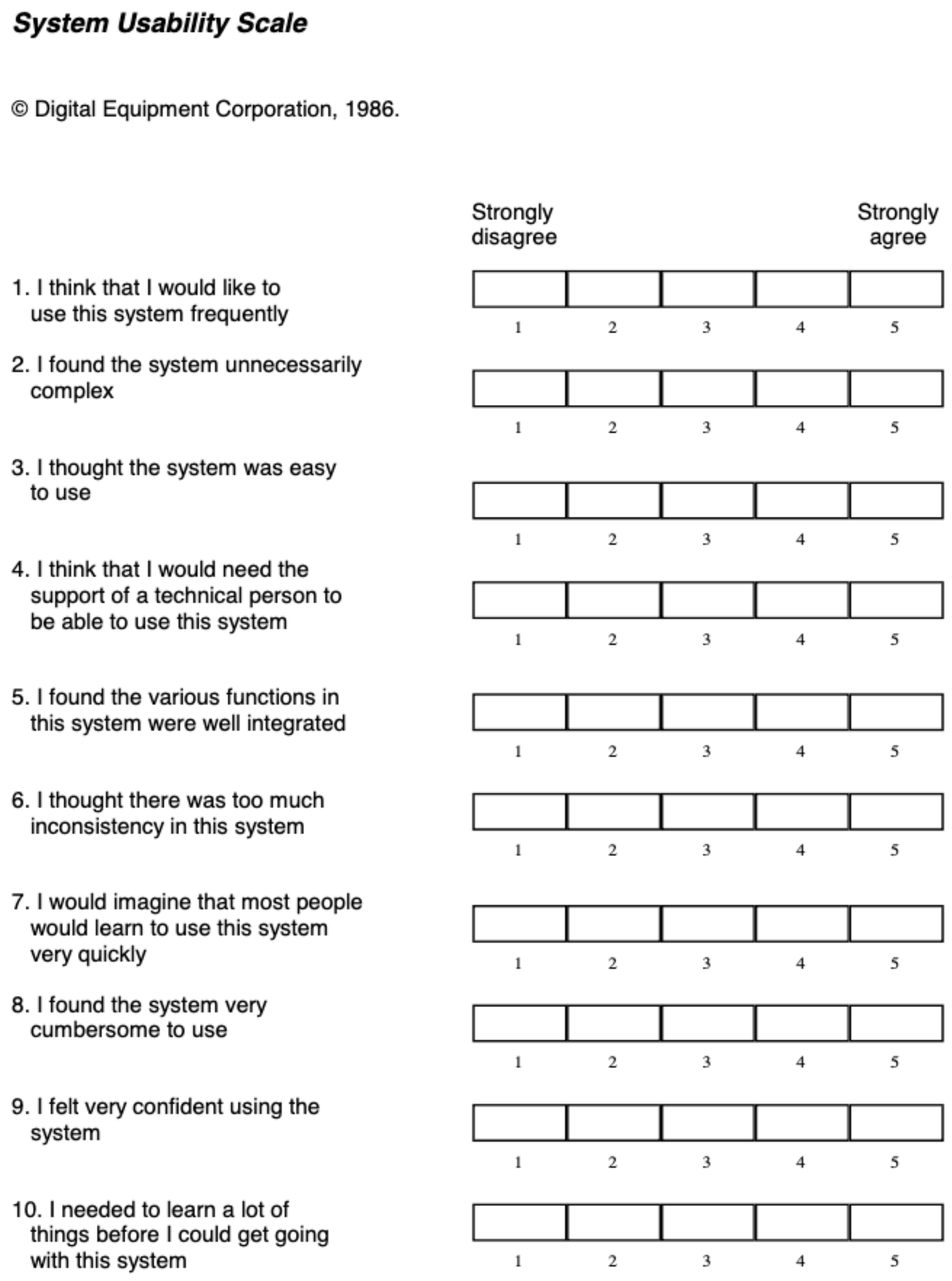
Physical Demand — How physically demanding was the task?
Very Low — Very High

Temporal Demand — How hurried or rushed was the pace of the task?
Very Low — Very High

Performance — How successful were you in accomplishing what you were asked to do?
Perfect — Failure

Effort — How hard did you have to work to accomplish your level of performance?
Very Low — Very High

Frustration — How insecure, discouraged, irritated, stressed, and annoyed wereyou?
Very Low — Very High



System Usability Scale

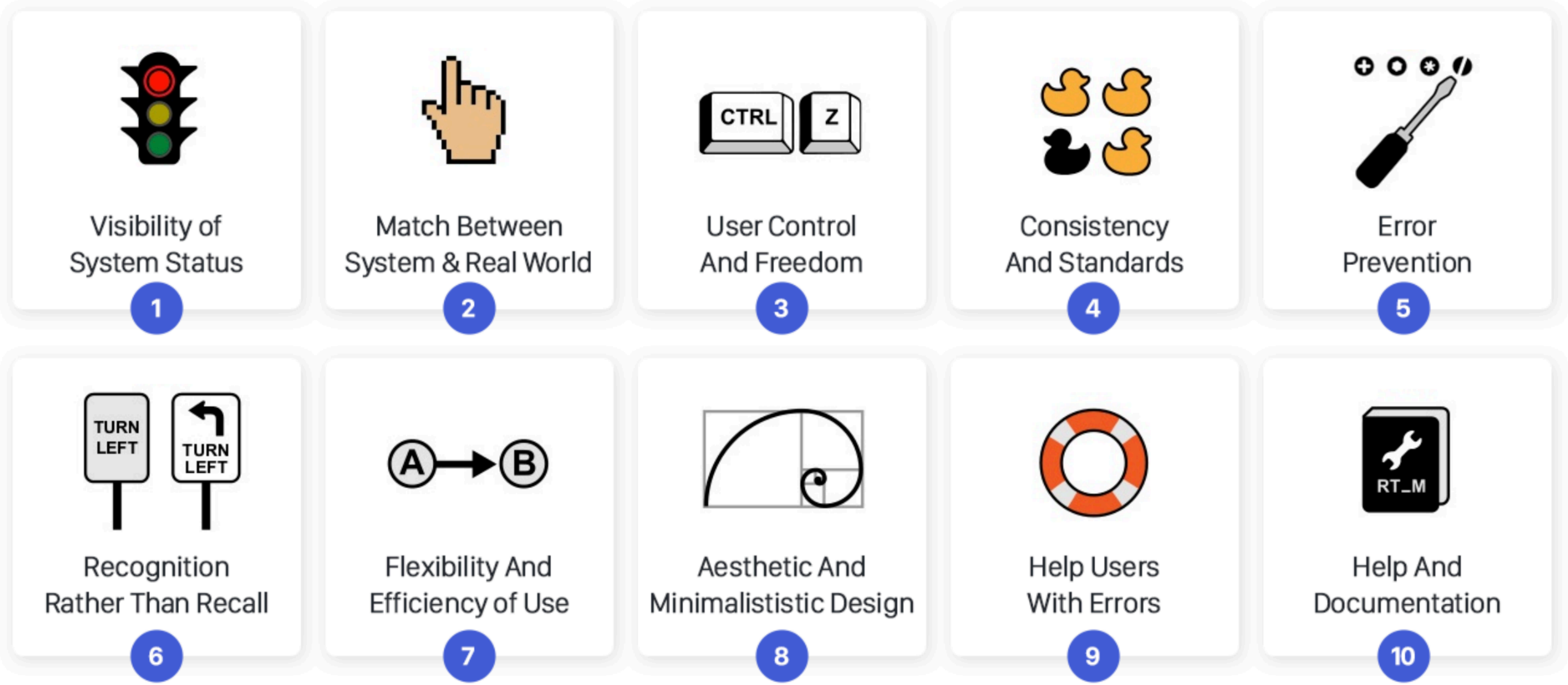
© Digital Equipment Corporation, 1986.

	Strongly disagree				Strongly agree
1. I think that I would like to use this system frequently	1	2	3	4	5
2. I found the system unnecessarily complex	1	2	3	4	5
3. I thought the system was easy to use	1	2	3	4	5
4. I think that I would need the support of a technical person to be able to use this system	1	2	3	4	5
5. I found the various functions in this system were well integrated	1	2	3	4	5
6. I thought there was too much inconsistency in this system	1	2	3	4	5
7. I would imagine that most people would learn to use this system very quickly	1	2	3	4	5
8. I found the system very cumbersome to use	1	2	3	4	5
9. I felt very confident using the system	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this system	1	2	3	4	5

Usability/UX evaluation with Standardized Questionnaires (SUS, NASA-TLX)

Evaluating the usability of a system

Analytical methods



Inostroza, Rodolfo, et al. "Usability heuristics for touchscreen-based mobile devices: update." Proceedings of the 2013 Chilean Conference on Human-Computer Interaction. 2013.

Nielsen's 10 usability heuristics

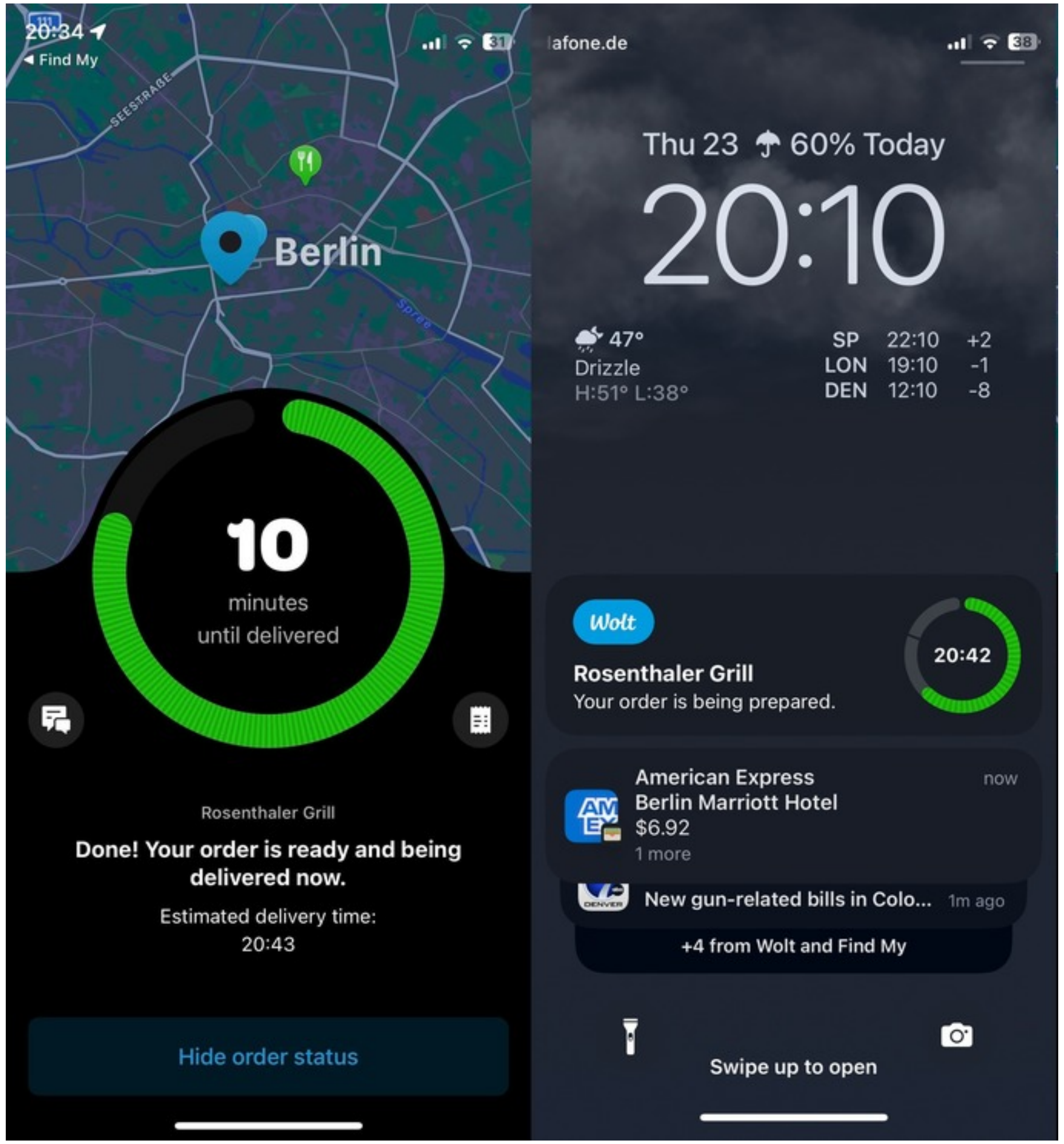
Human-centered design practice vs. HCI research

The HCD process and usability evaluations are great tools for addressing *specific* problems, in a *specific* context, and for *specific* user groups.

- The goal is to build a good, usable product
- Example: Running a usability test on Wolt's delivery status screen and notifications

HCI research is concerned with understanding how to design interactive systems in ways that are useful *across* contexts, beyond the scope of a specific problem and user group.

- Goal: Generate knowledge that informs the design of future systems and interaction patterns.
- Example: Studying how interruptions affect mobile interaction to develop generalizable principles for designing notification systems across apps and devices.



Wolt

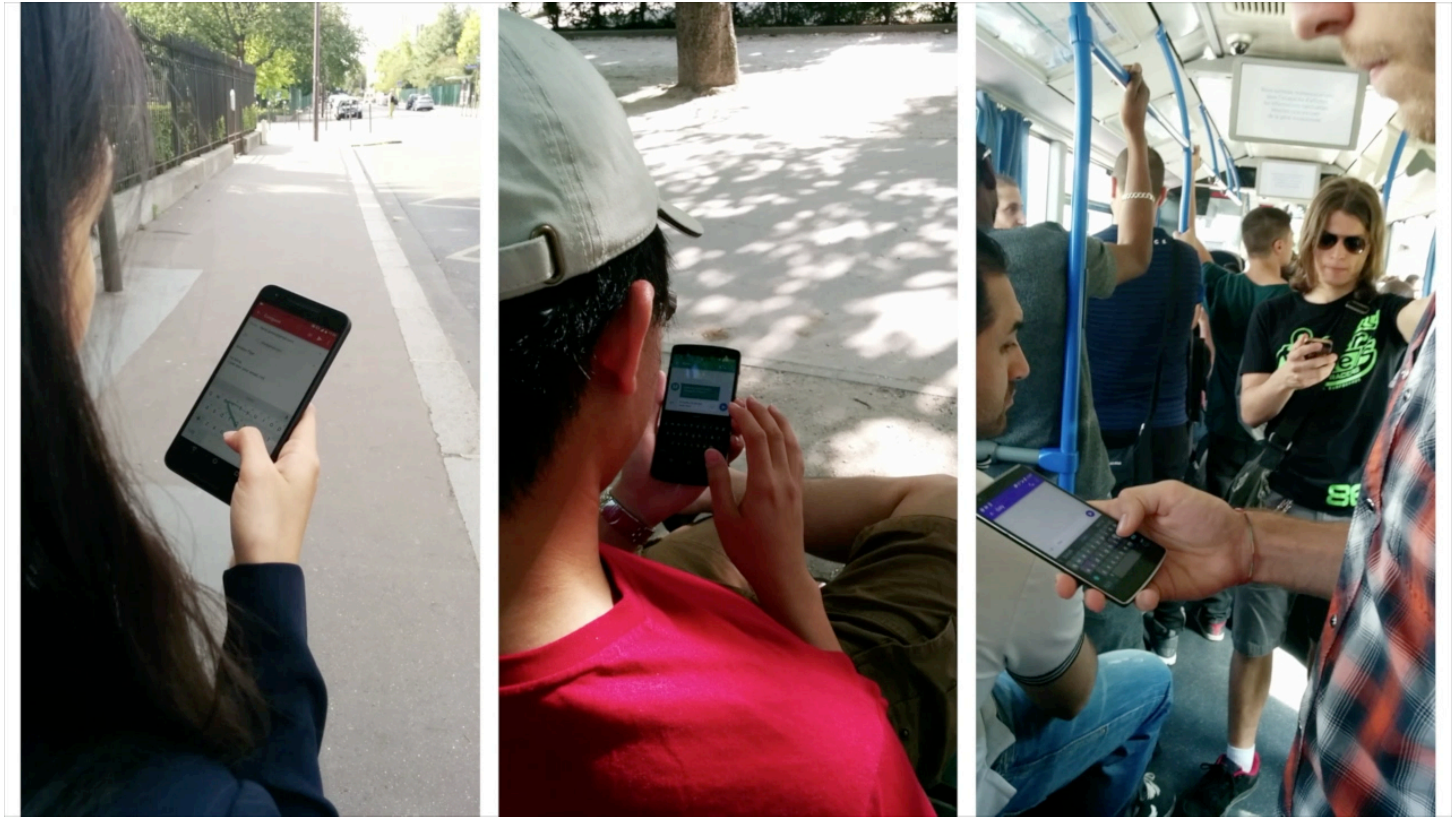
Usability (practice) vs. HCI (research)

Why should you care about this distinction?

Human-centered design practice vs. HCI research

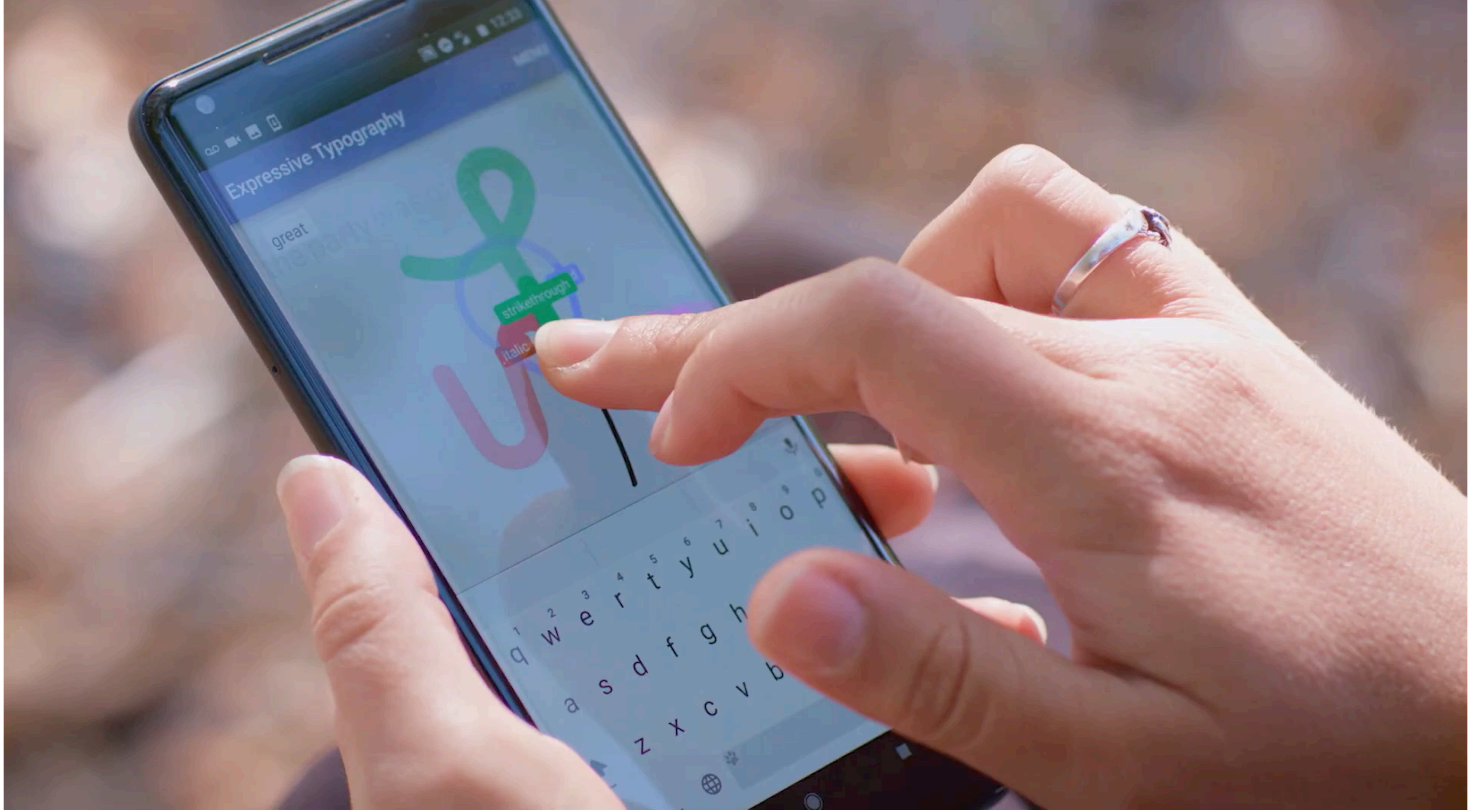
Why should you care?

- Understanding the SoTA in HCI expands your view of “what's even possible"
- A research mindset leads to innovation (e.g., patents)
- It’s a way of shaping the future (5-10 years ahead) instead of only solving current problems



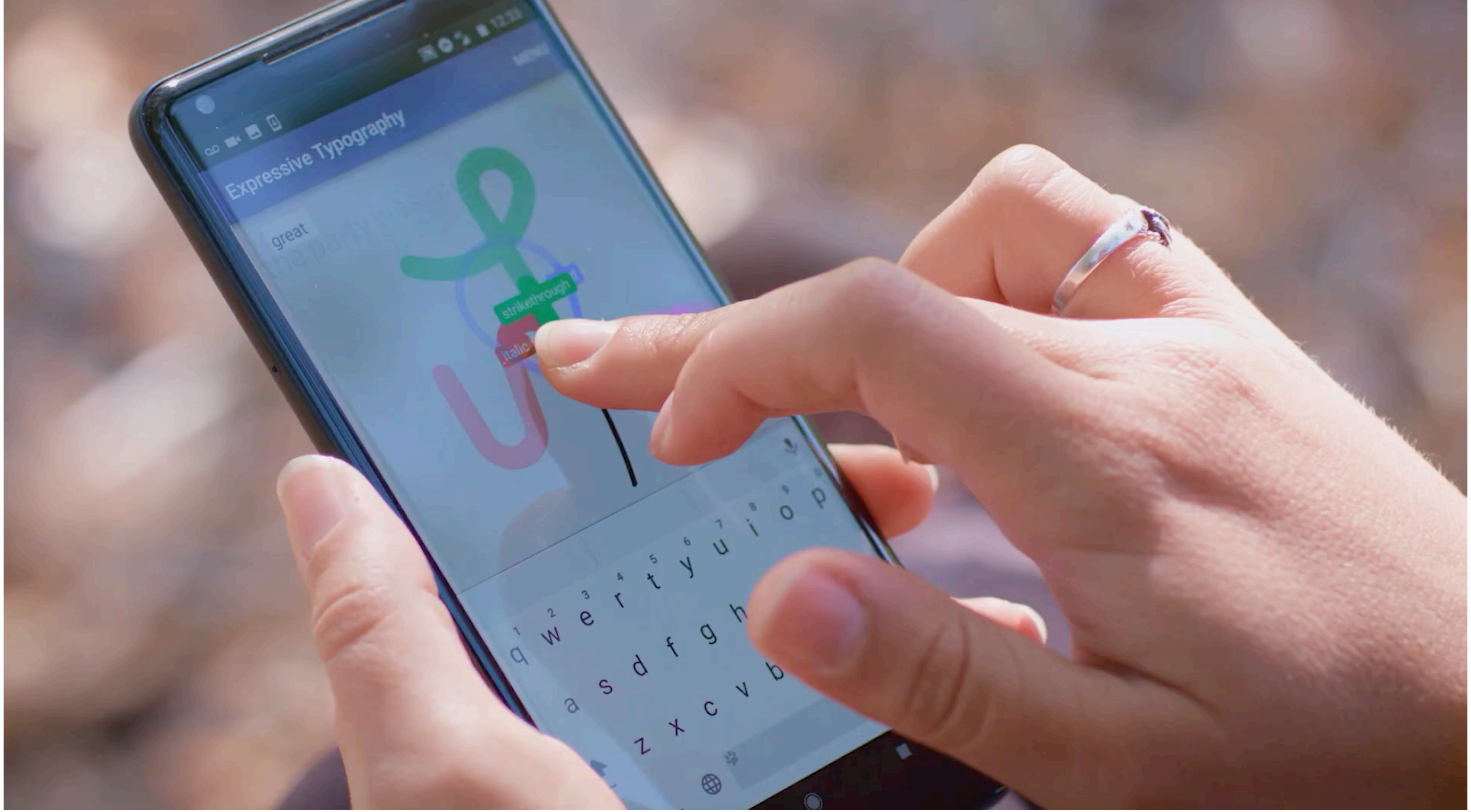
While HCD practice adapts to constraints, HCI research challenges constraints ("what if it was different?")

Jessalyn Alvina, Joseph Malloch, and Wendy E. Mackay. 2016. Expressive Keyboards: Enriching Gesture-Typing on Mobile Devices. In Proceedings of the 29th Annual Symposium on User Interface Software and Technology (UIST '16). Association for Computing Machinery, New York, NY, USA, 583–593. https://doi.org/10.1145/2984511.2984560



Example of patent resulting from research

Jessalyn Alvina, Carla F. Griggio, Xiaojun Bi, and Wendy E. Mackay. 2017. CommandBoard: Creating a General-Purpose Command Gesture Input Space for Soft Keyboard. In Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology (UIST '17). Association for Computing Machinery, New York, NY, USA, 17–28. https://doi.org/10.1145/3126594.3126639



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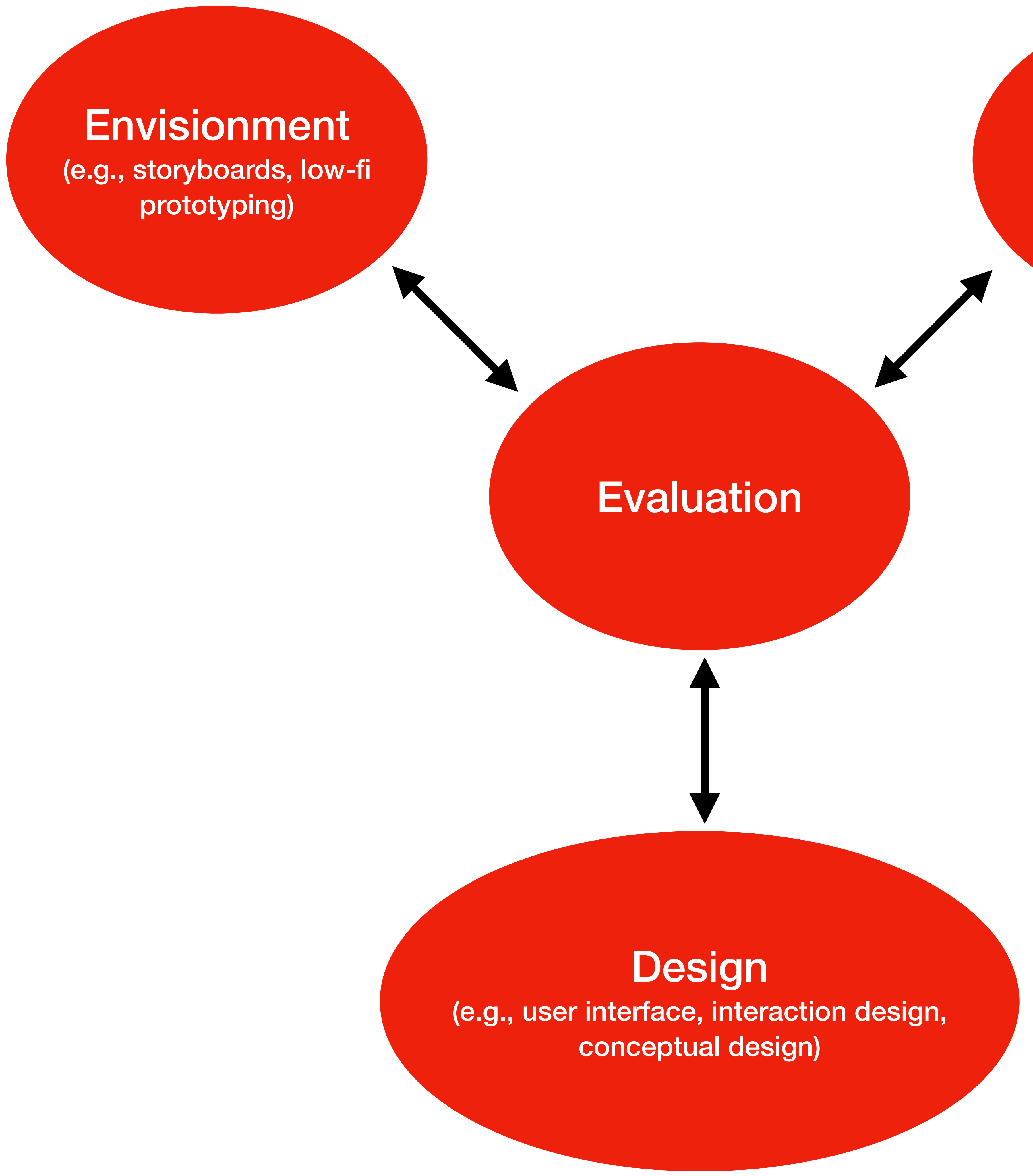
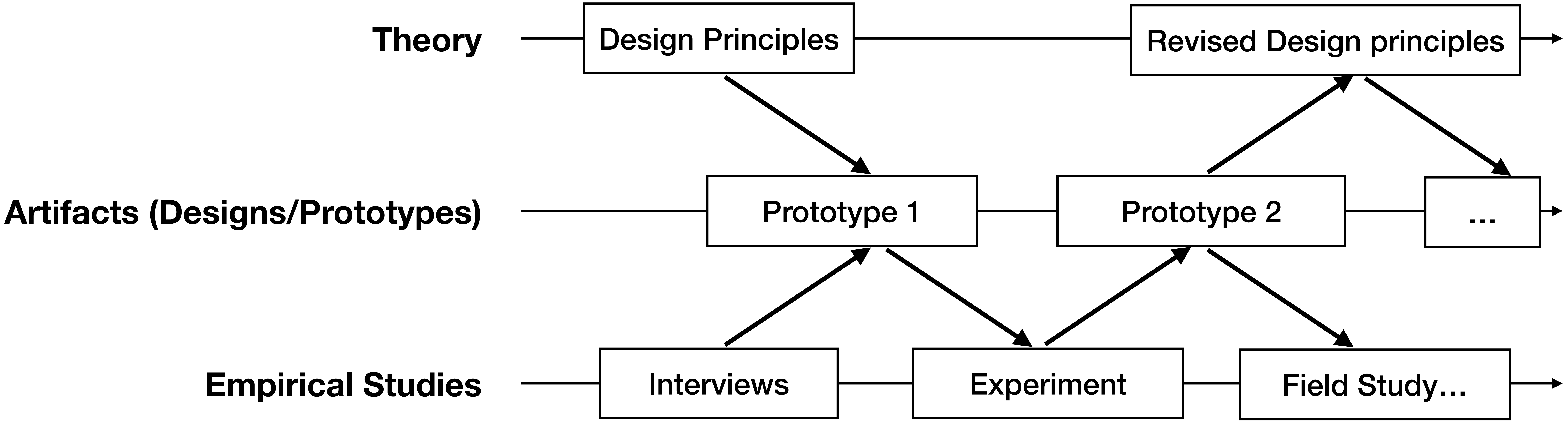
Human-centred design process of interactive systems



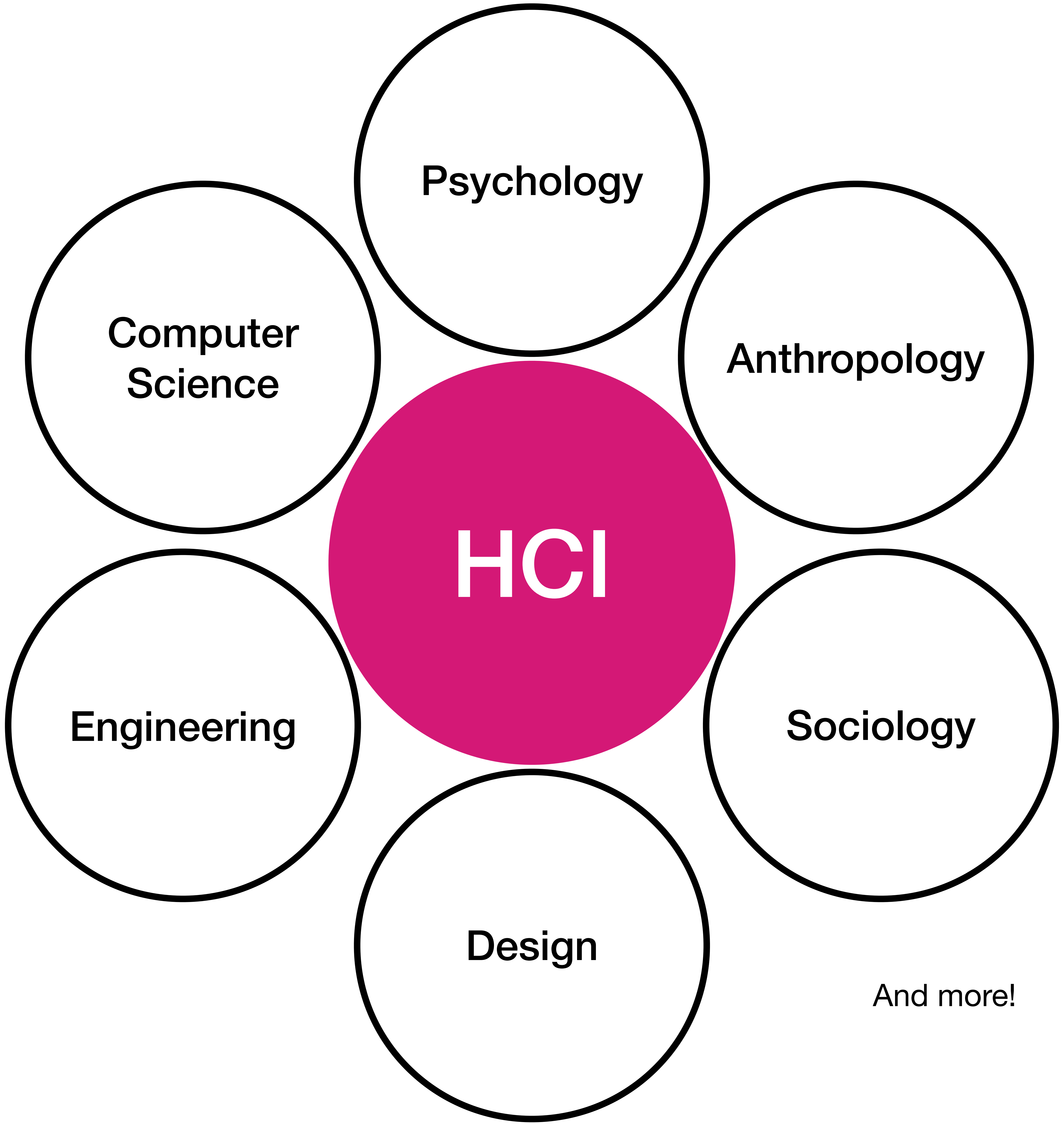
David Benyon, 3rd edition, "Designing User Experience"

HCI research process



HCI, Natural Science and Design: A Framework for Triangulation Across Disciplines
Wendy E. Mackay & Anne-Laure Fayard
https://www.lri.fr/~mackay/pdffiles/DIS97.Triangulate.pdf

HCI as a multi-disciplinary field



What kinds of contributions (output) are common in HCI research?

1. Empirical research contributions

Interviews, Surveys (questionnaires), Diary Studies, Field Studies, Experiments…

Emoji Accessibility for Visually Impaired People

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ABSTRACT
Emoji are graphical symbols that appear in many aspects of our lives. Worldwide, around 36 million people are blind and 217 million have a moderate to severe visual impairment. This portion of the population may use and encounter emoji, yet it is unclear what accessibility challenges emoji introduce. We first conducted an online survey with 58 visually impaired participants to understand how they use and encounter emoji online, and the challenges they experience. We then conducted 11 interviews with screen reader users to understand more about the challenges reported in our survey findings. Our interview findings demonstrate that technology is both an enabler and a barrier, emoji descriptors can hinder communication, and therefore the use of emoji impacts social interaction. Using our findings from both studies, we propose best practice when using emoji and recommendations to improve the future accessibility of emoji for visually impaired people.

and are used by politicians and government bodies [36, 55], travel companies [54], media outlets, and public figures (e.g., singer Katy Perry who has one of the largest Twitter followings [51]). Emoji have even been discussed within official court transcripts [35], and resulted in convictions [23].

People interpret emoji differently, and emoji design variations across different platforms (e.g., iOS vs Android) can exacerbate misunderstandings [45, 64]. Furthermore, emoji are often used beyond their original intended meaning, which adds another layer of complexity to disambiguating the intended use of an emoji [64, 74]. Prior research on emoji has largely focused on those with typical vision. However, it is estimated that 36 million people worldwide are blind and 217 million have a moderate to severe visual impairment [73]. Prior work highlighted challenges visually impaired people face when using technology [7] and social media [22, 49]. However, it is not clear what accessibility challenges occur with emoji.

Poor Use in Context: Our participants highlighted that emoji used in different contexts can lead to specific challenges. Decorative emoji, e.g. emoji in usernames on social media, caused challenges as many decorative emoji could be announced by a screen reader. An example of this is shown in Figure 2.A.

> P7: *"Try listening to 'cat with heart shaped eyes fireworks sparkles watermelon kissing face flag of Andorra' a few times in a row and you get the frustration."*

Visual Design: Emoji design also caused challenges for 22% of the participants. For participants who had some residual vision, this was often related to the use of colour such as P6 who described that *"the colors of the heart [emoji] can be too similar."*. For blind participants, differences between design of the visual emoji and the description were challenging:

> P28: *"Some emoji [are] useless or just have a bad design (I was told the 'pray' emoji [🙏] is actually a 'high five')."*

Misunderstanding: This relates to the use of visual representations of things that blind users had not experienced. This sometimes made it difficult to select an emoji.

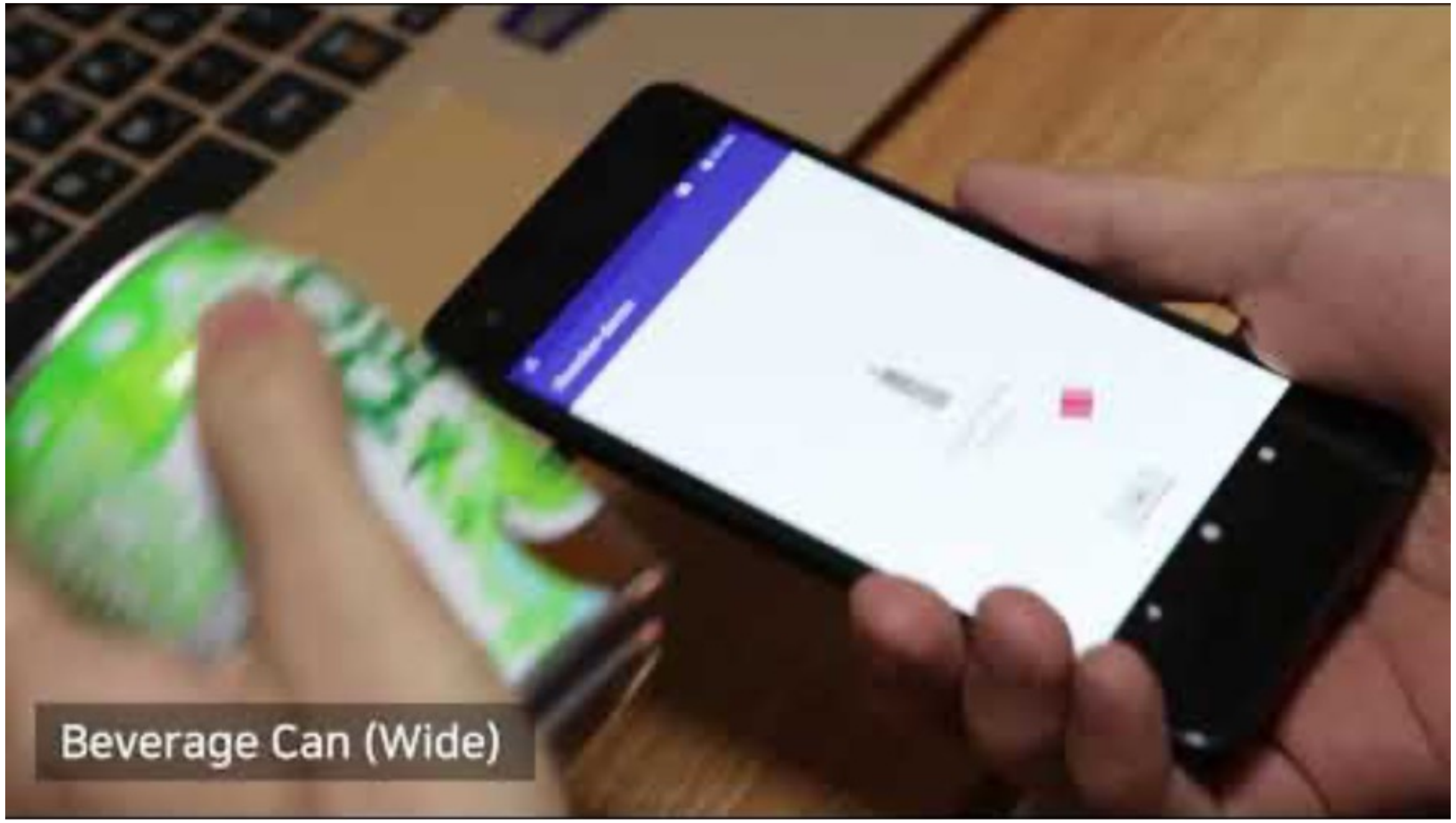
> P38: *"...I entered the word 'happy', and it suggested many faces, which were all described to me; however, as I have never had vision, I was unable to know which face was the most appropriate for my situation."*

Semi-structured Interviews

Surveys (questionnaires)

2. Artifact contributions

Systems, Input devices, Hardware, Interaction Technique, …



Interaction technique + system

Taesik Gong, Hyunsung Cho, Bowon Lee, and Sung-Ju Lee. 2019. Knocker: Vibroacoustic-based Object Recognition with Smartphones. Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. 3, 3, Article 82 (September 2019), 21 pages. https://doi.org/10.1145/3351240

2. Artifact contributions

Hardware, Interaction Techniques

Mathieu Le Goc, Lawrence H. Kim, Ali Parsaei, Jean-Daniel Fekete, Pierre Dragicevic, and Sean Follmer. 2016. Zooids: Building Blocks for Swarm User Interfaces. In Proceedings of the 29th Annual Symposium on User Interface Software and Technology (UIST '16). Association for Computing Machinery, New York, NY, USA, 97–109. https://doi.org/10.1145/2984511.2984547

3. Methodological contributions

New standardized questionnaires, new methods for collecting data, etc.

Scale	#	Final User Burden Scale Question Set and Subscales: Currently using (present tense)	Final User Burden Scale Question Set and Subscales: Abandoned (past tense)	Response type
Difficulty of Use	1	I need assistance from another person to use [X].	I needed assistance from another person to use [X].	1
	2	[X] demands too much mental effort.	[X] demanded too much mental effort.	1
	3	It takes too long for me to do what I want to do with [X].	It took too long for me to do what I wanted to do with [X].	1
	4	[X] is hard to learn.	[X] was hard to learn.	2
Physical	5	Using [X] too much creates physical discomfort.	Using [X] too much created physical discomfort.	2
	6	[X] has made me feel physical pain.	[X] had made me feel physical pain.	1
	7	Use of [X] is too physically demanding.	Use of [X] was too physically demanding.	1
Time and Social	8	I spend too much time using [X].	I spent too much time using [X].	2
	9	I use [X] more often than I should.	I used [X] more often than I should have.	1
	10	[X] distracts me from social situations.	[X] distracted me from social situations.	1
	11	Using [X] has a negative effect on my social life.	Using [X] had a negative effect on my social life.	1
Mental and Emotional	12	[X] requires me to remember too much information.	[X] required me to remember too much information.	1
	13	[X] presents too much information at once.	[X] presented too much information at once.	1
	14	Using [X] makes me feel like a bad person.	Using [X] made me feel like a bad person.	1
	15	I feel guilty when I use [X].	I felt guilty when I used [X].	1
Privacy	16	I am worried about what information gets shared by [X].	I was worried about what information got shared by [X].	2
	17	[X]'s policies about privacy are not trustworthy.	[X]'s policies about privacy were not trustworthy.	2
	18	[X] requires me to do a lot to maintain my privacy within it.	[X] required me to do a lot to maintain my privacy within it.	1
Financial	19	[X] is too expensive.	[X] was too expensive.	2
	20	The upfront cost to using [X] is too high.	The upfront cost to using [X] was too high.	2

User Burden Scale: A tool for assessing user burden in computing systems.

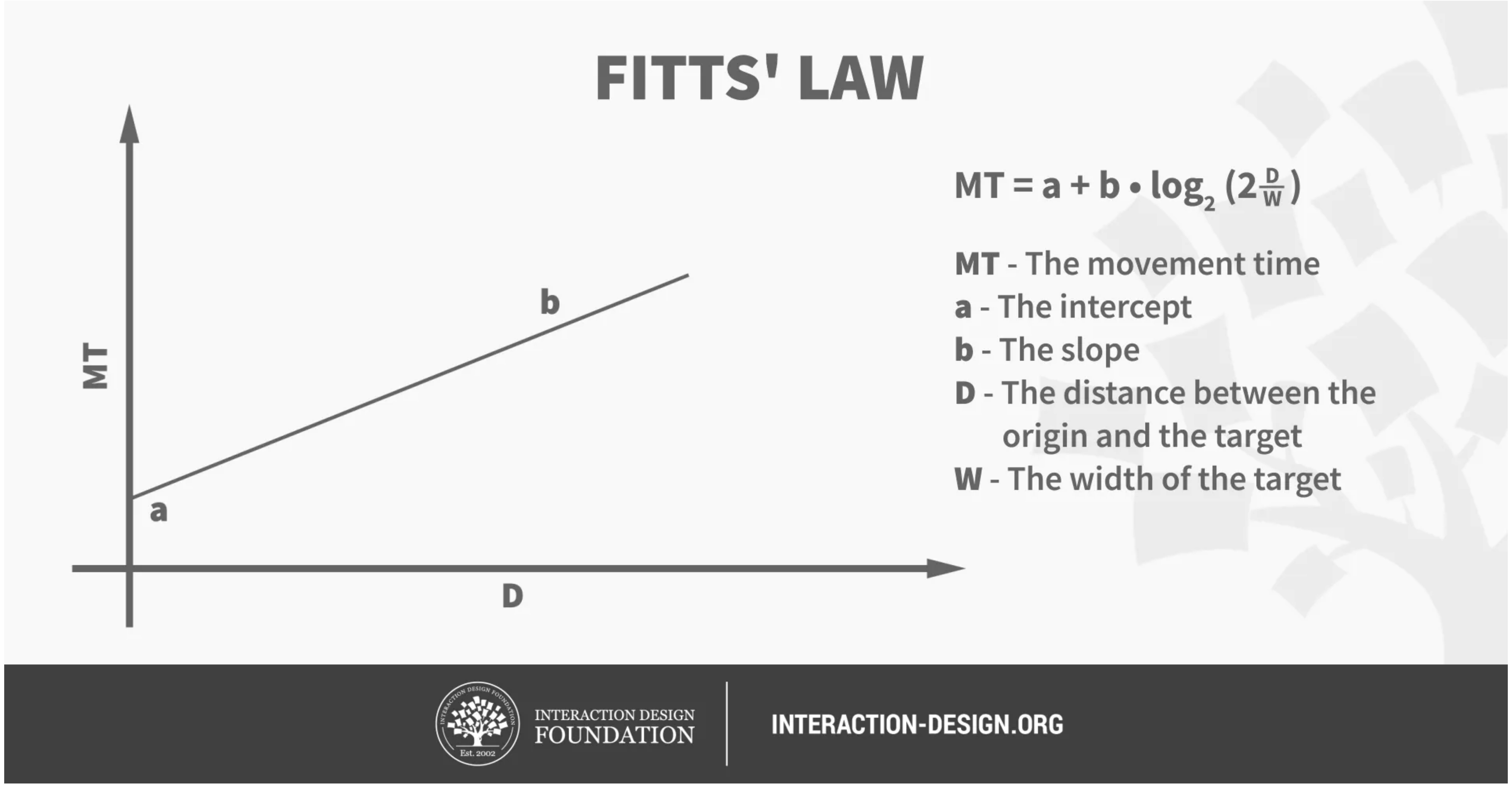
Model for user burden that consists of six constructs of "burden": 1) difficulty of use, 2) physical, 3) time and social, 4) mental and emotional, 5) privacy, and 6) financial.

New standardized questionnaire

Suh, H., Shahriaree, N., Hekler, E.B., and Kientz, J.A. Developing and validating the User Burden Scale: A tool for assessing user burden in computing systems. Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI '16). ACM Press, New York. http:// dx.doi.org/10.1145/2858036.2858448

4. Theoretical Contributions

Concepts, Models, Frameworks, Descriptive / Predictive / Generative theories, Design Spaces

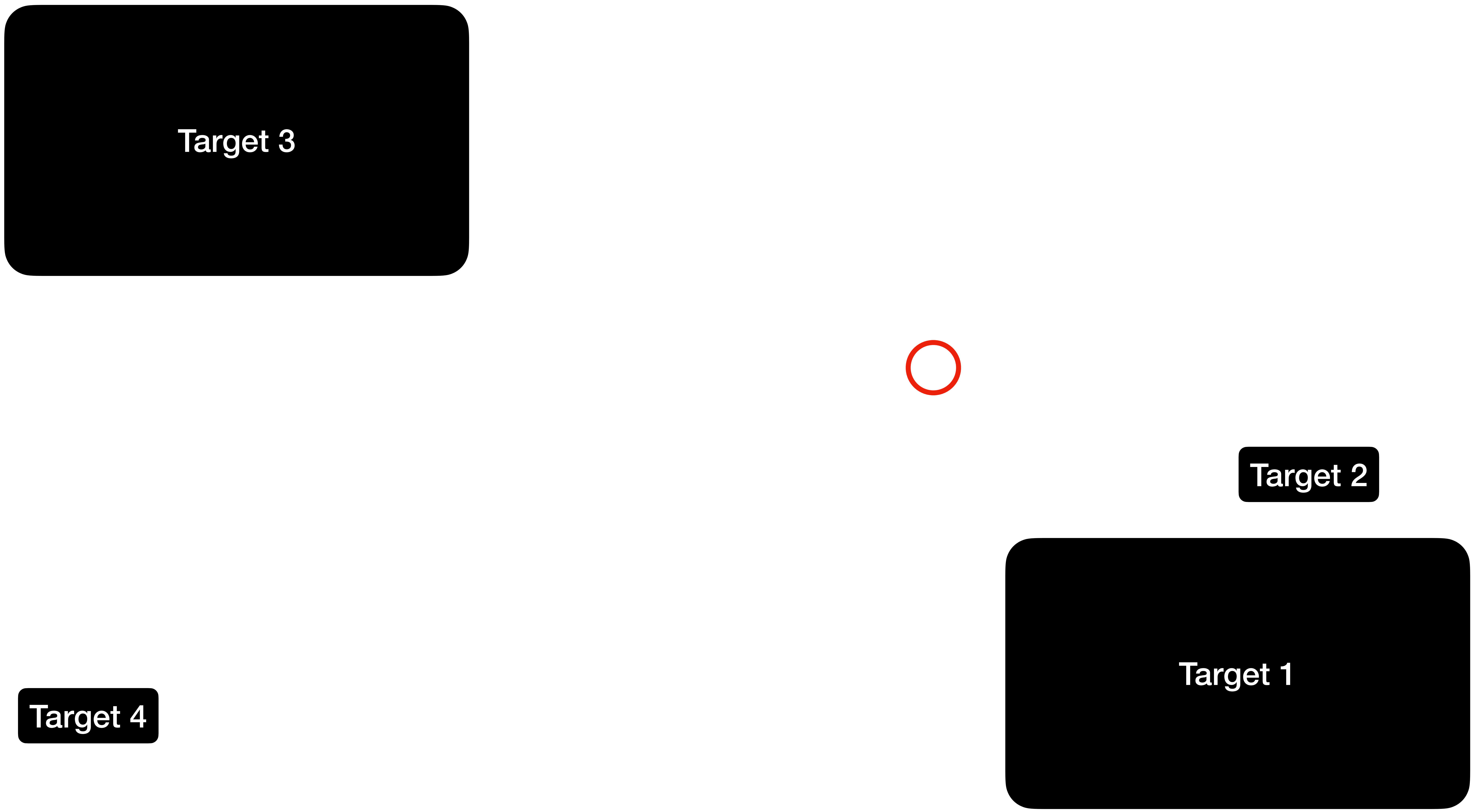


Predictive Model

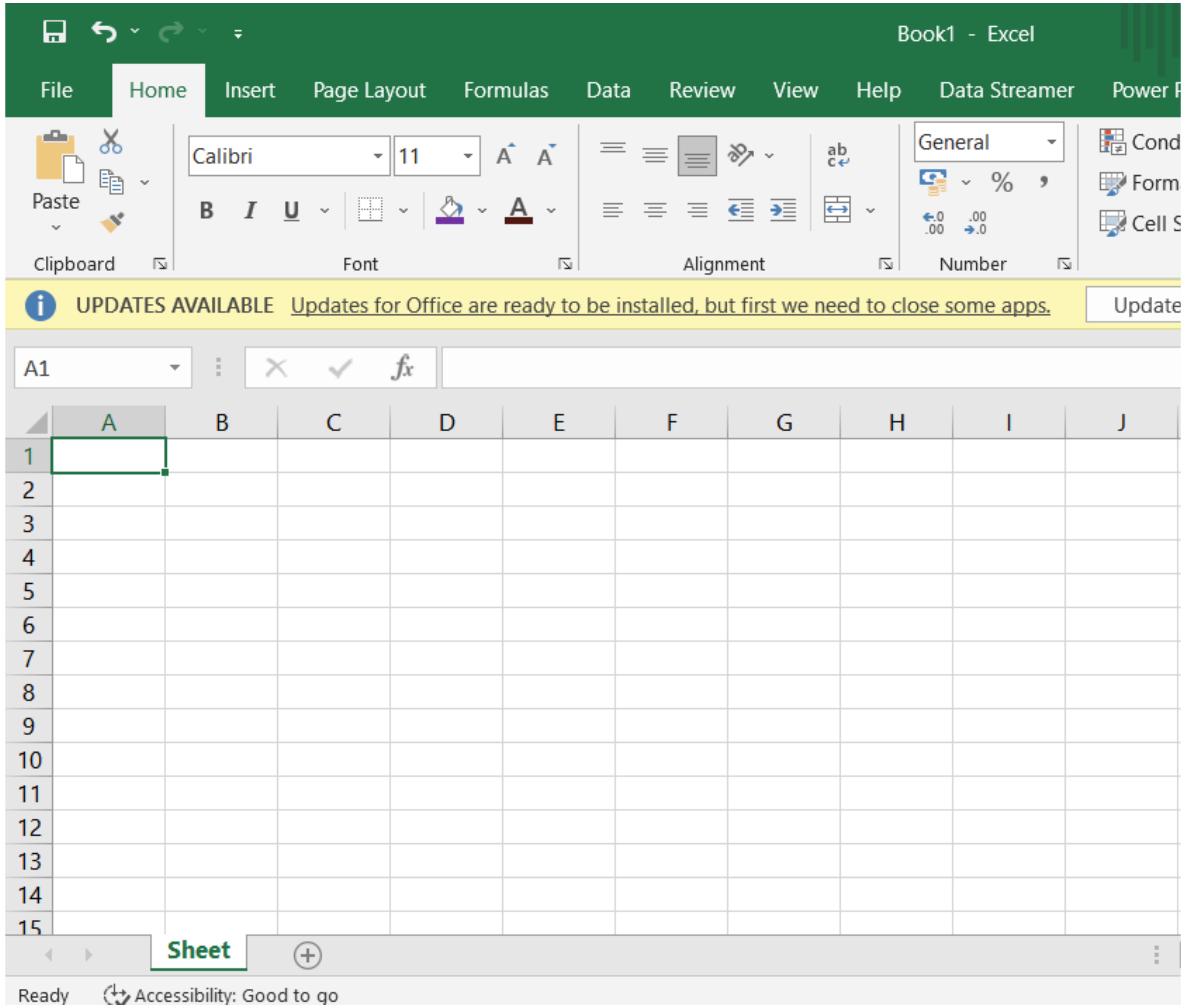
Fitts' law

https://www.interaction-design.org/literature/article/fitts-s-law-the-importance-of-size-and-distance-in-ui-design

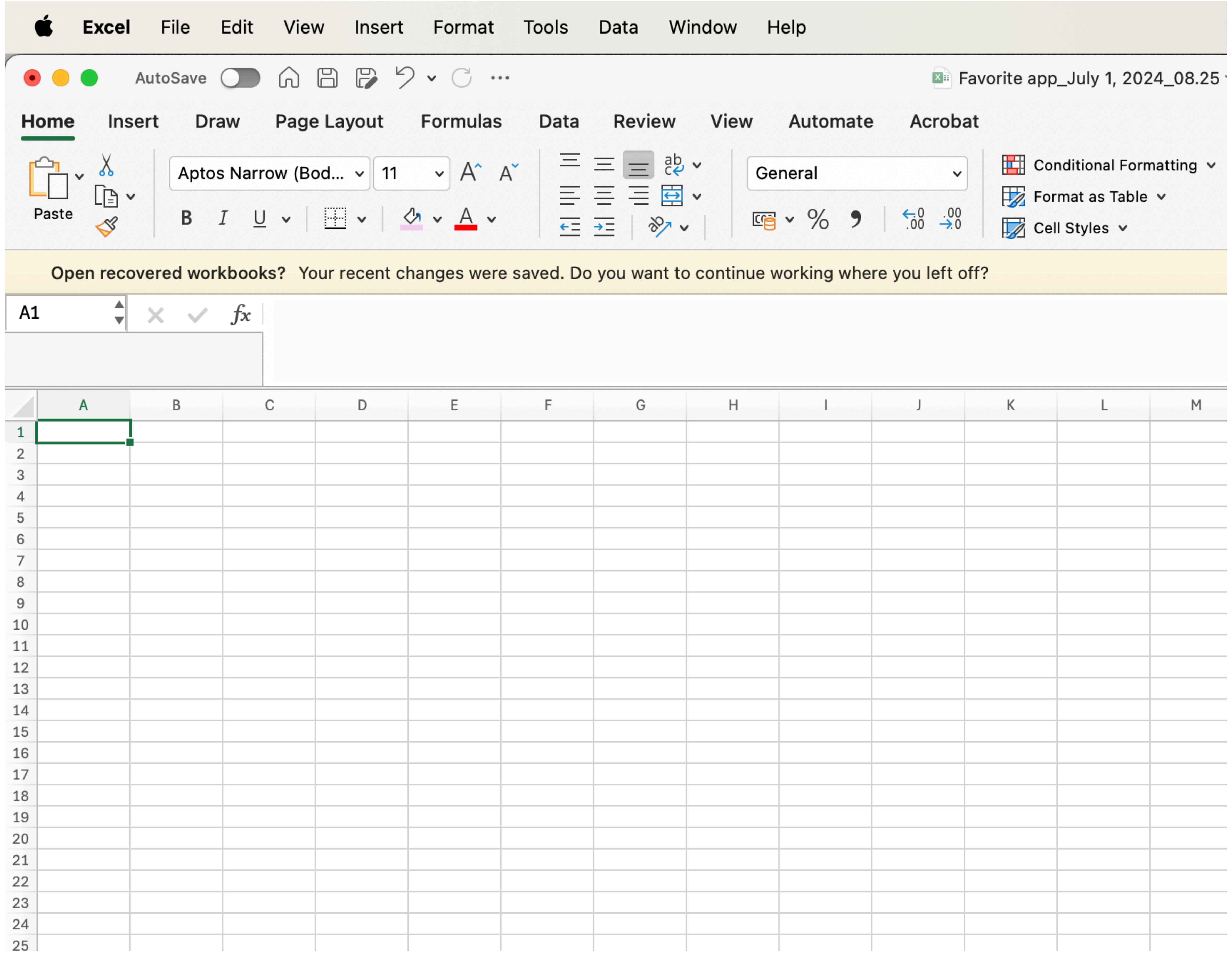
Fitts' law



Fitts' law



Windows



Mac

5. Dataset Contributions

Concepts, Models, Frameworks, Descriptive / Predictive / Generative theories, Design Spaces

K-EmoPhone, a **real-world** smartphone and wearable dataset with *in-situ* emotion, stress, and attention labels acquired from **77 students over seven days**. This dataset aims to contribute to:

(1) understanding human affects with behavioral, contextual, and physiological data,
(2) obtaining fine-grained affect states in terms of time, and
(3) utilization in multiple domains, ranging from affective computing to attention management.

Participants had to use:

- Their Android smartphones (for collecting location, app usage, movement)
- Microsoft Band 2 smartwatch (for collecting acceleration, step count, distance, ambient light, UV exposure, heart rate)

They also had to respond to ESM (Experience Sampling Methods) questionnaires about how they felt on the spot.

Data set: https://doi.org/10.5281/zenodo.7606611

Kang, S., Choi, W., Park, C.Y. *et al.* K-EmoPhone: A Mobile and Wearable Dataset with *In-Situ* Emotion, Stress, and Attention Labels. *Sci Data* **10**, 351 (2023). https://doi.org/10.1038/s41597-023-02248-2

6. Survey (Literature Review) Contributions

Analysis and synthesis of previous research on a topic / research area

Synchronization category includes 55% of the papers, and communication includes 46% of them. Moreover, storage access includes 40% of the papers. The category with the least amount of papers was “intelligent” systems with only 4 papers (5% of the total amount of included documents).

Classification based on the Evaluation Method

Several types of evaluation methods were applied and described in the papers included in this SLR. Table 2 shows the distribution of papers by evaluation method applied.

Evaluation Method	Papers	Percentage
Case Study	[2,5,7,9,11,14,22,25,26,37,60,65,69, 78,83,103]	19,2%
Empirical Evaluation	[8]	1,3%
Example	[4,21,36,102,106]	6%
Experiment	[32,77,94,95,98]	6%
Heuristics	[67]	1,3%
Usability	[1,24,27,33,43,73,107]	8,4%
Performance Evaluation	[101]	1,3%
Preliminary Study (Lab)	[17,19,31,35,50,62,70,75,79,87,90,9 1]	14,4%
Not Specified	[6,8,10,13,15,16,18,20,28,34,42,45, 46,48,51,55–57,63,64,68,72,74,80–82,85,86,88,89,92,96,99,100,104]	42%

Table 2. Distribution of papers by evaluation method applied.

Classification based on data gathering and notification mechanisms

A limited amount of research has been focused on the study of notification mechanisms applied in CSCW systems. Figure 4 shows the data gathering and notification mechanisms described in the papers included in this SLR. The data gathering mechanisms showed in Figure 4 include:

- Light: A lamp is used to gather information; users touch the lamp to signal an event.
- Table top: A display mounted in a table is used to gather commands from the users.
- User controls: Refers to remote controls or devices provided to the users to interact with the system.
- Mobile: Smartphones, tablets, and PDAs.
- Sensors: Devices embedded in the user environment or place in their body to gather information, users did not interact directly with the sensors.
- Eye-tracker: Device that measures the point of gaze and the motion of an eye relative to the head.
- Traditional Interfaces: Laptops, computers and traditional mechanisms of interaction (i.e., mouse, keyboard, screen)
- Vibro-tactile belt: A wearable device that includes sensors that allows the user interaction with the system and data gathering.

The notification mechanisms showed in Figure 4 include:

- Light: is used to display information to the users, for instance the intensity of a light notifies the users that their talking time during a meeting is ending.
- Mobile: Phone messages or calls, sounds or vibration are used as notification mechanisms of events.
- Public Display: Large wall mounted displays are located in the collaboration contexts to allow multiple users to see public information.

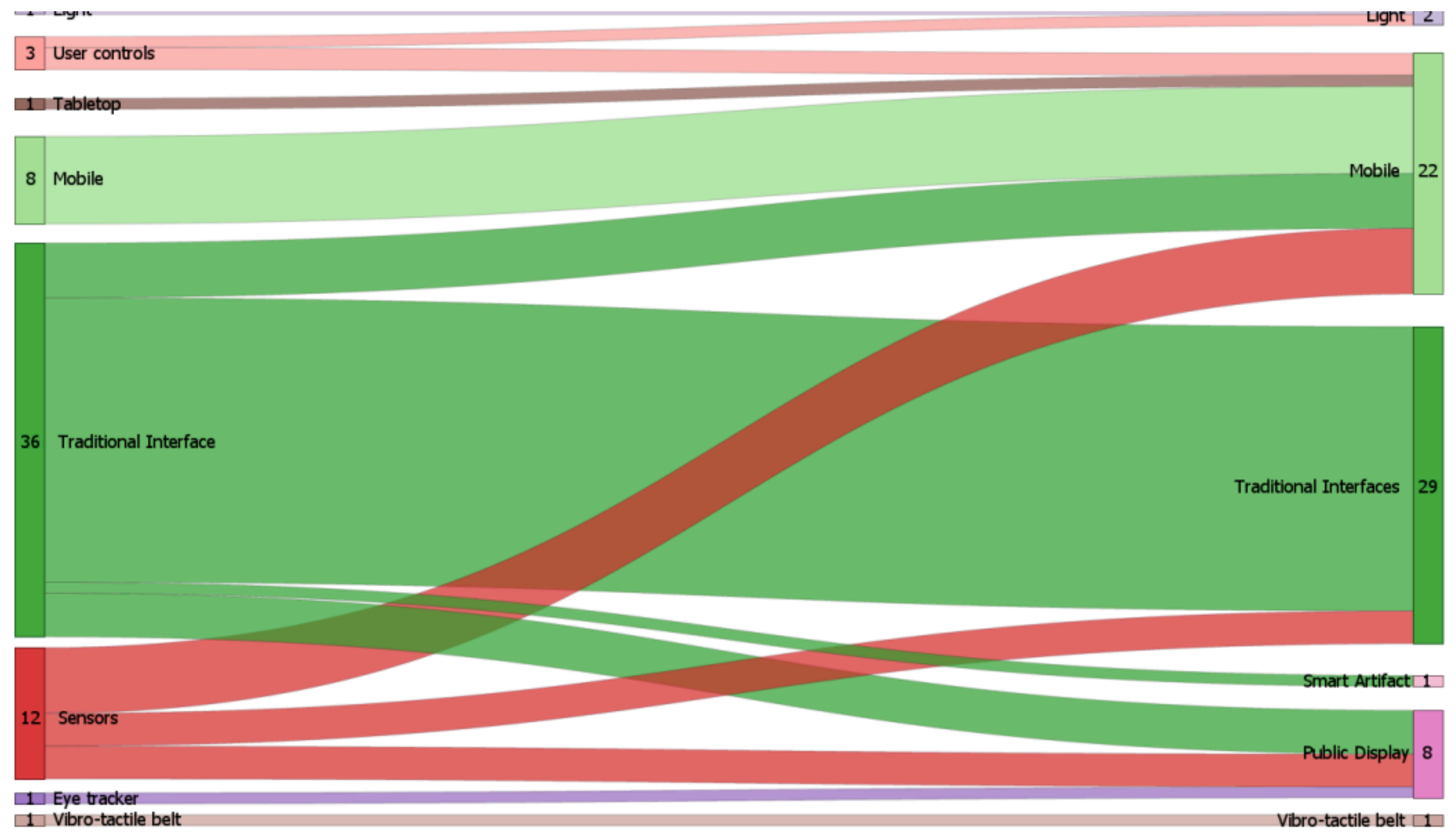


Figure 4. Sankey diagram visualizing the relation between data gathering mechanisms (left) and notification mechanisms (right).

Gustavo Lopez and Luis A. Guerrero. 2017. Awareness Supporting Technologies used in Collaborative Systems: A Systematic Literature Review. In Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing (CSCW '17). Association for Computing Machinery, New York, NY, USA, 808–820. https://doi.org/10.1145/2998181.2998281

HCI “concerns” evolved through time!

Three waves of HCI research

	First wave
Main concerns	Human performance, efficiency, error reduction
View of the user	Information processor, individual operator
Typical context	Lab, workstation, task-focused settings
Main methods	Controlled experiments, usability tests
View of generalization	Universal laws, replicable findings
Example mobile question	How fast/accurately can users tap targets while walking?

Three waves of HCI research

	First wave	Second wave
Main concerns	Human performance, efficiency, error reduction	Interaction in social & organizational context
View of the user	Information processor, individual operator	Social actor embedded in practices
Typical context	Lab, workstation, task-focused settings	Workplaces, groups, real-world settings
Main methods	Controlled experiments, usability tests	Field studies, ethnography, participatory design
View of generalization	Universal laws, replicable findings	Transferable insights within similar contexts
Example mobile question	How fast/accurately can users tap targets while walking?	How do messaging apps support family coordination?

Three waves of HCI research

	First wave	Second wave	Third wave
Main concerns	Human performance, efficiency, error reduction	Interaction in social & organizational context	Experience, meaning, culture, everyday life
View of the user	Information processor, individual operator	Social actor embedded in practices	Experiencing, meaning-making person
Typical context	Lab, workstation, task-focused settings	Workplaces, groups, real-world settings	Homes, public spaces, personal & social life
Main methods	Controlled experiments, usability tests	Field studies, ethnography, participatory design	In-the-wild studies, cultural probes, research-through-design, qualitative + mixed methods
View of generalization	Universal laws, replicable findings	Transferable insights within similar contexts	Situated understanding, conceptual or theoretical transfer
Example mobile question	How fast/accurately can users tap targets while walking?	How do messaging apps support family coordination?	How do smartphones shape people's sense of presence or wellbeing?

Let’s go back to our examples.
How do you see each wave reflected?

So what makes Mobile HCI special?

Exercise: Recap on Nielsen's Heuristics

How do these heuristics apply to mobile contexts?

1. Get in groups of 2-3 members
2. Choose a heuristic (let's try to cover them all!) - see link on Moodle
3. Choose a mobile app / operating system / device at hand
4. Find 2-3 examples of a mobile interaction / UI that **follow** this heuristic
5. Find 1-2 examples that **violate** this heuristic
6. Reflect with your group:
    1. What are the challenges unique to mobile interactions that are not explicitly considered in these heuristics?
    2. How would you re-interpret / re-explain these heuristics to better fit mobile interactions?

7. **Copy** and edit the first 2 slides for your examples/reflection: https://aaudk-my.sharepoint.com/:p:/g/personal/wb25qs_cs_aau_dk/IQCRXW8ERQWZQLkhCvvyE9rpAfhN_7zYrj31SAgLRNAnksM?e=pLfanH

Challenges (Benyon)

Chapter 19 (Mobile Computing) in **Designing interactive systems: A comprehensive guide to HCI, UX and interaction design** by Benyon, David. Pearson, 2014.

Because of the small screen it is difficult to achieve the design principle of visibility.

Functions have to be tucked away and accessed by multiple levels of menu, leading to difficulties of navigation.

Another feature is that there is not room for many buttons, so each button has to do a lot of work. This results in the need for different ‘modes’ and this makes it difficult to have clear control over the functions.

Visual feedback is often poor and people have to stare into the device to see what is happening.

Thus other modalities such as sound and touch are often used to supplement the visual.